Appendix Part 1: Structurally Obfuscated DSP Circuits

DSP Applications – Data Flow Graphs, Scheduling and Datapath Circuits

1) 4-point DCT: Obfuscation

Equations of 4-point DCT:

X[0]=c2*x[0]+ c2*x[1]+ c2*x[2]+ c2*x[3]

X[1]=c1*x[0]+ c3*x[1]+ (-c3)*x[2]+ (-c1)*x[3]

X[2]=c2*x[0]+ (-c2)*x[1]+ (-c2)*x[2]+ c2*x[3]

X[3]=c3*x[0]+ (-c1)*x[1]+ c1*x[2]+ (-c3)*x[3]

The generic equation of 4-Point DCT to compute 1st sample:

X[0]=k1*x[0]+ k2*x[1]+ k3*x[2]+ k4*x[3]

DFG of 4-point DCT after Tree Height Transformation based obfuscation using (Sengupta & Roy et.al 2017) is shown here. Based on generic representation to compute 1st sample, the respective data flow graph is shown in Figure 1:

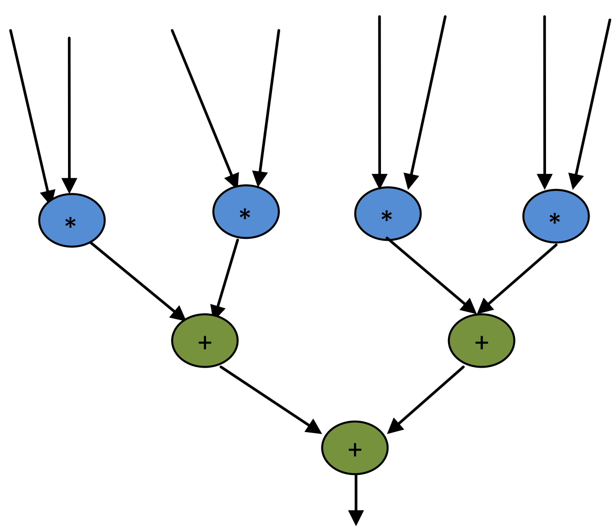


Figure 1

Scheduled DFG of 4-point DCT based on Resource constraint (1M,1A) after Tree Height Transformation based obfuscation:

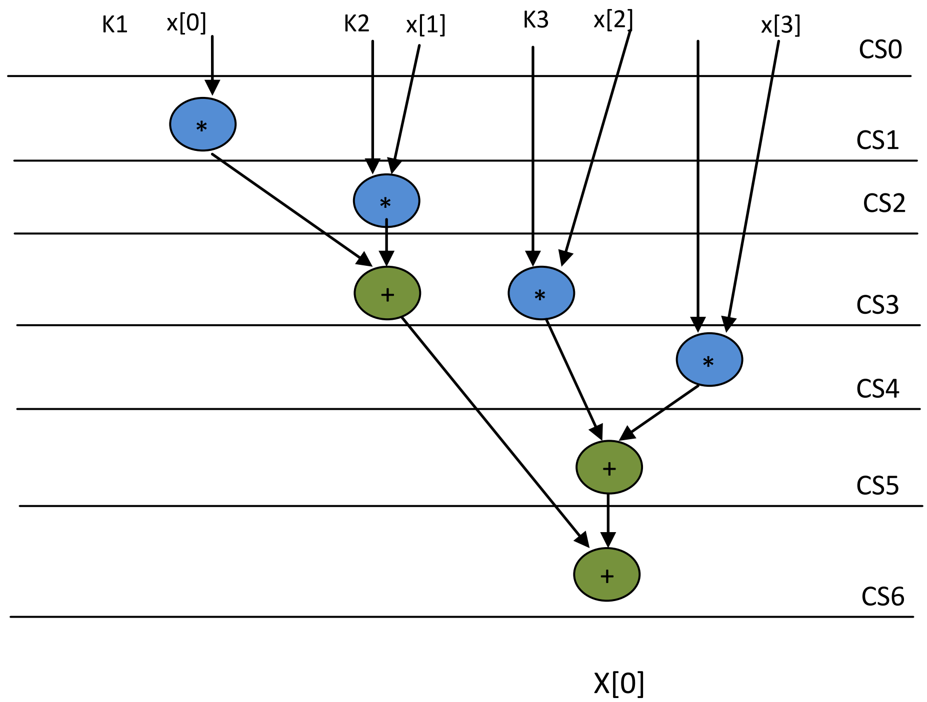


Figure 2

Equation of 4-Point DCT to compute 1st sample: X[0]=k1*x[0]+ k2*x[1]+ k3*x[2]+ k4*x[3]

RTL Datapath after Tree Height Transformation based obfuscation:

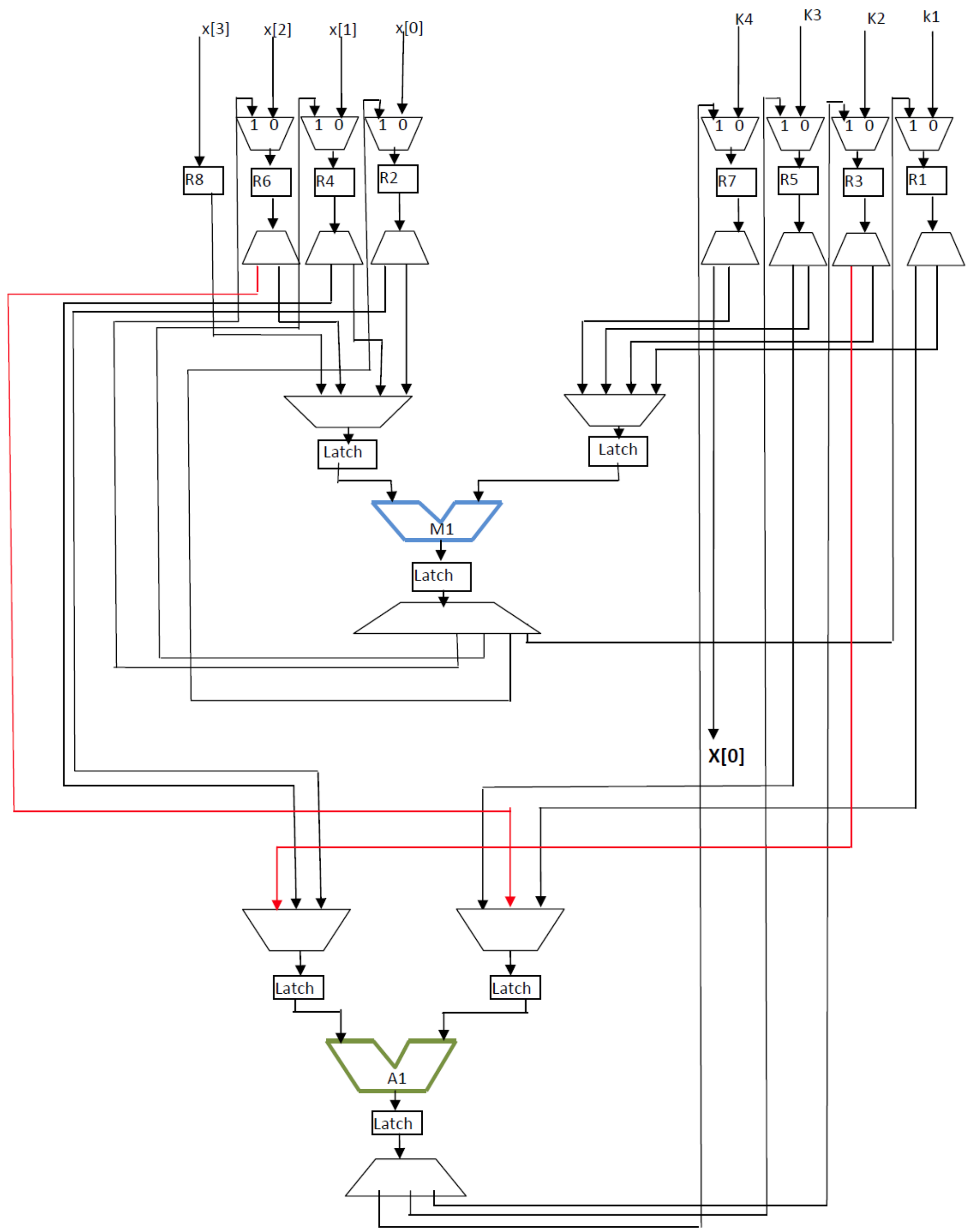


Figure 3

2) 8-point DCT: Obfuscation

X[0]=c4*x[0]+c4*x[1]+c4*x[2]+c4*x[3]+c4*x[4]+ c4*x[5]+ c4*x[6]+ c4*x[7]

X[1]=c1*x[0]+c3*x[1]+c5*x[2]+c7*x[3]+(-c7)*x[4]+(- c5)*x[5]+ (-c3)*x[6]+ (-c1)*x[7]

X[2]=c2*x[0]+c6*x[1]+(-c6)*x[2]+(-c2)*x[3]+(-c2)*x[4]+(- c6)*x[5]+ c6*x[6]+ c2*x[7]

X[3]=c3*x[0]+(-c7)*x[1]+(-c1)*x[2]+(-c5)*x[3]+c5*x[4]+ c1*x[5]+ c7*x[6]+ (-c3)*x[7]

X[4]=c4*x[0]+(-c4)*x[1]+(-c4)*x[2]+c4*x[3]+c4*x[4]+ (-c4)*x[5]+ (-c4)*x[6]+ c4*x[7]

X[5]=c5*x[0]+(-c1)*x[1]+c7*x[2]+c3*x[3]+(-c3)*x[4]+ (-c7)*x[5]+ c1*x[6]+ (-c5)*x[7]

X[6]=c6*x[0]+(-c2)*x[1]+c2*x[2]+(-c6)*x[3]+(-c6)*x[4]+ c2*x[5]+ (-c2)*x[6]+ c6*x[7]

X[7]=c7*x[0]+(-c5)*x[1]+c3*x[2]+(-c1)*x[3]+c1*x[4]+ (-c3)*x[5]+ c5*x[6]+ (-c7)*x[7]

Generic representation of 8-Point DCT to compute 1st sample:

X[0]=k1*x[0]+ k2*x[1]+ k3*x[2]+ k4*x[3]+ k5*x[4]+ k6*x[5]+ k7*x[6]+ k8*x[7]

DFG of 8-Point DCT after Tree Height Transformation:

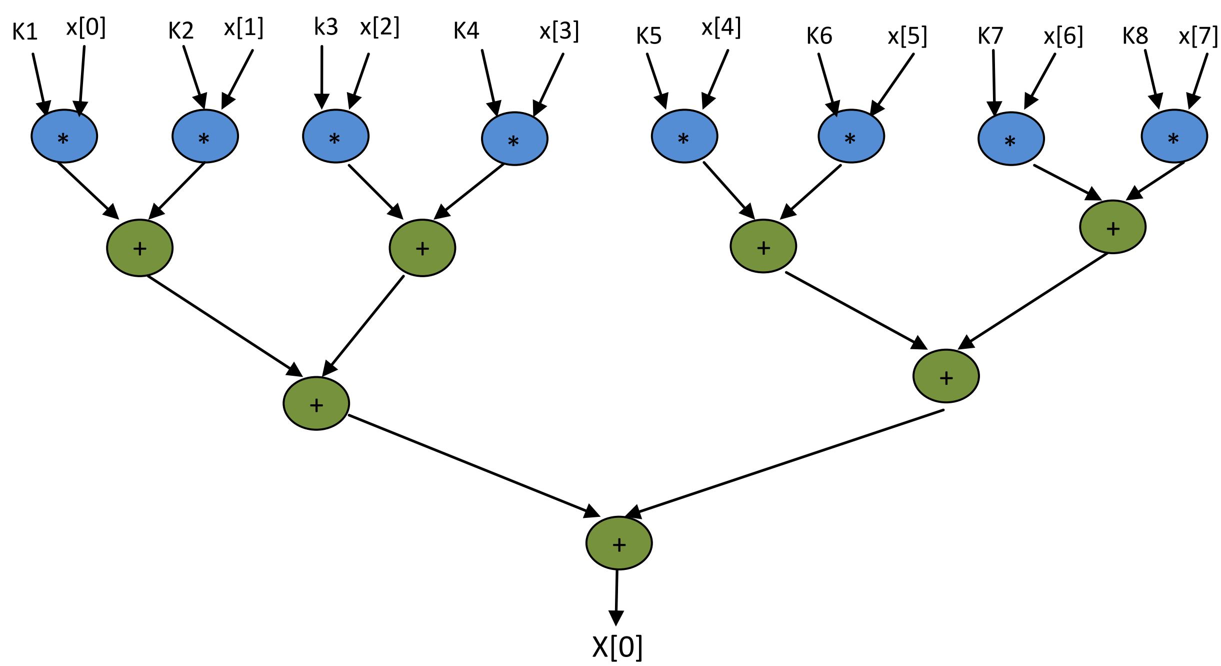


Figure 4

Scheduled DFG of 8-point DCT after Tree Height Transformation based on Resource constraints (1A, 1M):

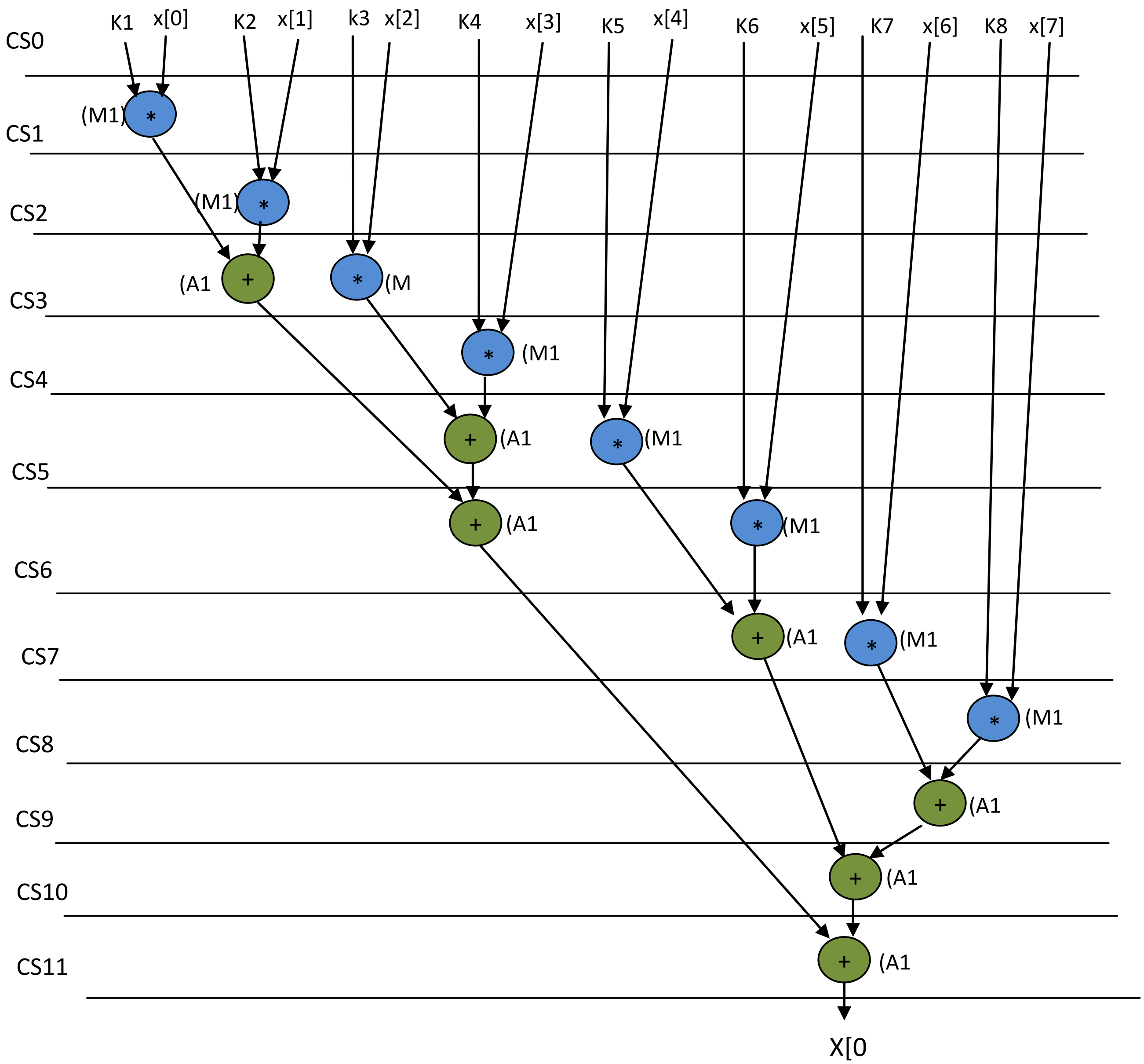


Figure 5

Equation of 8-Point DCT: X[0]=k1*x[0]+ k2*x[1]+ k3*x[2]+ k4*x[3]+ k5*x[4]+ k6*x[5]+ k7*x[6]+ k8*x[7]

RTL Datapath after Tree Height Transformation based obfuscation:

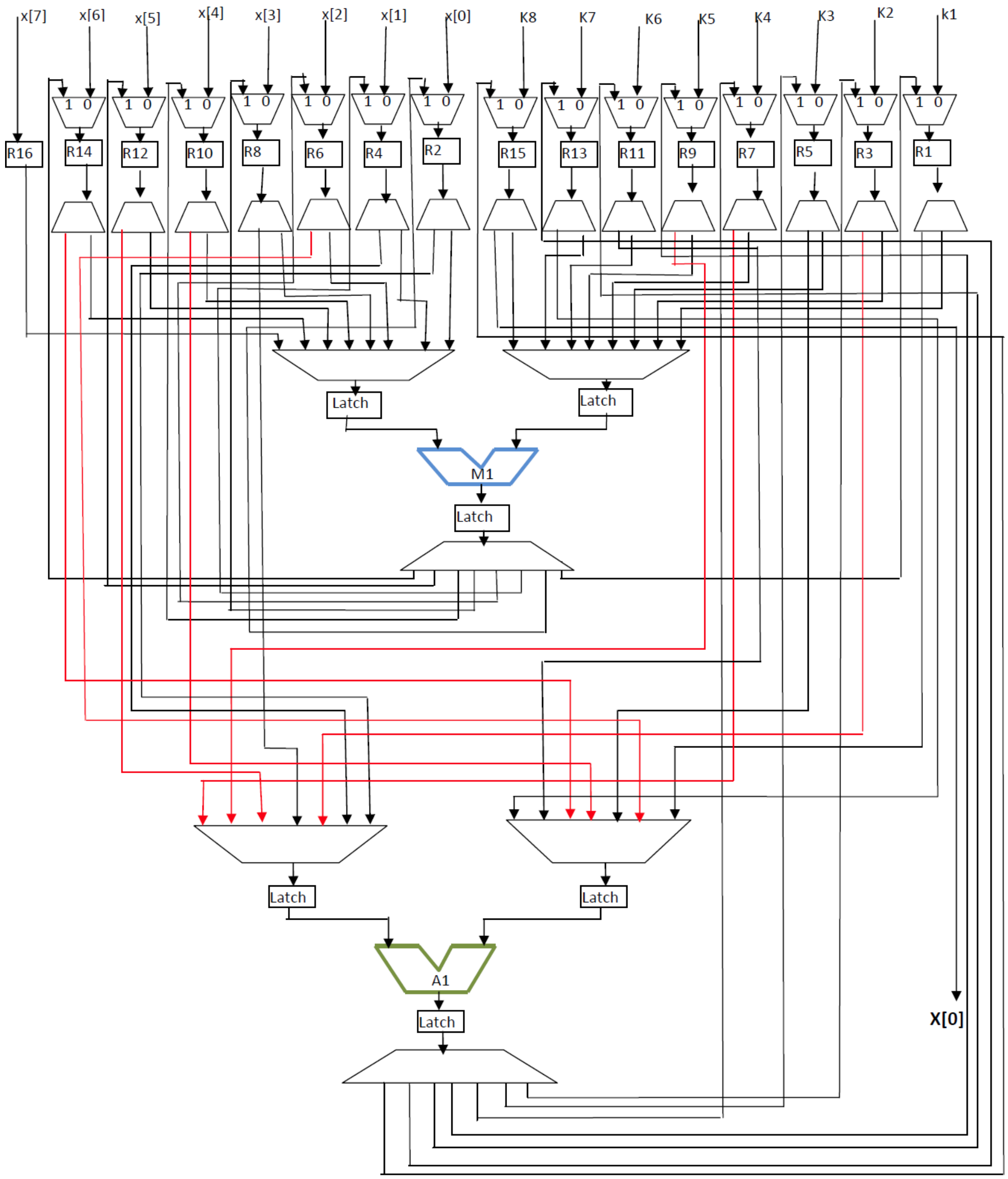


Figure 6

3) 8-point IDCT: Obfuscation

Equations of 8-point IDCT:

x’[0]=c4*X’[0]+c1*X’[1]+c2*X’[2]+c3*X’[3]+c4*X’[4]+ c5*X’[5]+ c6*X’[6]+ c7*X’[7]

x’[1]=c4*X’[0]+c3*X’’[1]+c6*X’[2]+(-c7)*X’[3]+(-c4)*X’[4]+(- c1)*X’[5]+ (-c2)*X’[6]+ (-c5)*X’[7]

x’[2]=c4*X’[0]+c5*X’[1]+(-c6)*X’[2]+(-c1)*X’[3]+(-C4)*X’[4]+ c7*X’[5]+ c2*X’[6]+ c3*X’[7]

x’[3]=c4*X’[0]+c7*X’[1]+(-c2)*X’[2]+(-c5)*X’[3]+c4*X’[4]+ c3*X’[5]+ (-c6)*X’[6]+ (-c1)*X’[7]

x’[4]=c4*X’[0]+(-c7)*X’[1]+(-c2)*X’[2]+c5*X’[3]+c4*X’[4]+ (-c3)*X’[5]+ (-c6)*X’[6]+ c1*X’[7]

x’[5]=c4*X’[0]+(-c5)*X’[1]+(-c6)*X’[2]+c1*X’[3]+(-c4)*X’[4]+ (-c7)*X’[5]+ c2*X’[6]+ (-c3)*X’[7]

x’[6]=c4*X’[0]+(-c3)*X’[1]+c6*X’[2]+c7*X’[3]+(-c4)*X’[4]+ c1*X’[5]+ (-c2)*X’[6]+ c5*X’[7]

x’[7]=c4*X’[0]+(-c1)*X’[1]+c2*X’[2]+(-c3)*X’[3]+c4*X’[4]+ (-c5)*X’[5]+ c6*X’[6]+ (-c7)*X’[7]

Generic representation of 8-Point IDCT to compute 1st sample:

x‘[0]=k’1*X’[0]+ k’2*X’[1]+ k’3*X’[2]+ k’4*X’[3]+ k’5*X’[4]+ k’6*X’[5]+ k’7*X’[6]+ k’8*X’[7]

DFG of 8-Point IDCT after Tree Height Transformation:

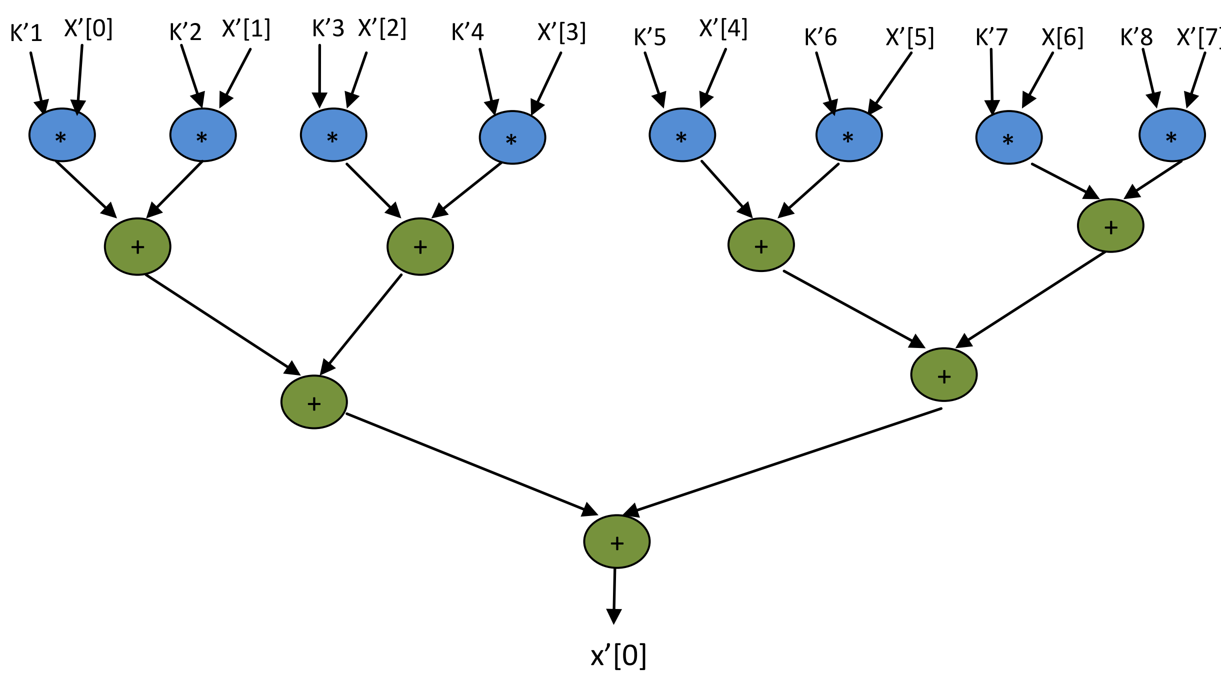


Figure 8.17

Scheduled DFG of 8-point IDCT after Tree Height Transformation based on Resource constraints (1A, 1M).

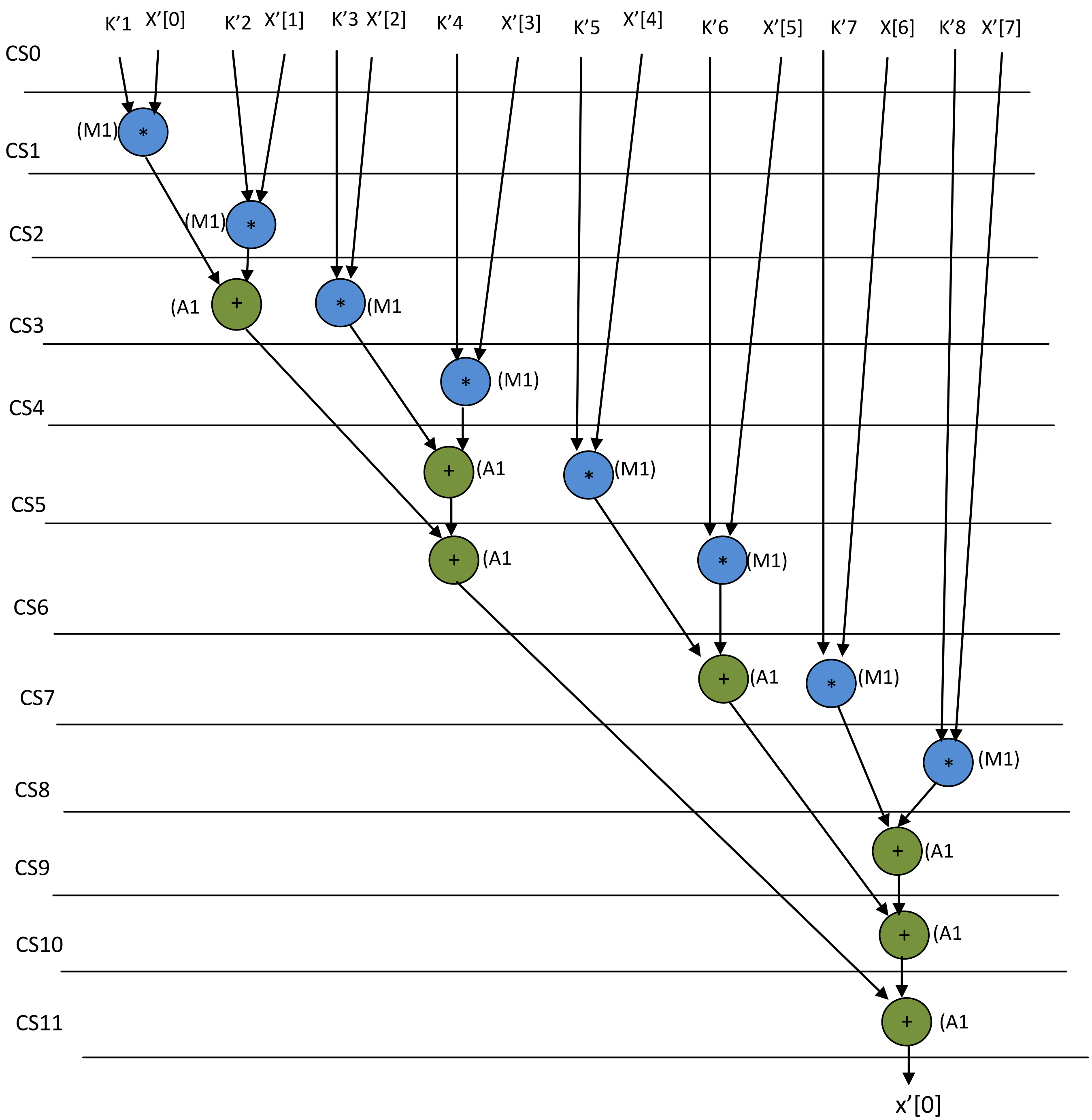


Figure 8.18

Equation of 8-Point IDCT: x'[0]=k'1*X'[0]+ k'2*X'[1]+ k'3*X'[2]+ k'4*X'[3]+ k'5*X'[4]+ k'6*X'[5]+ k'7*X'[6]+ k'8*X'[7]

Datapath after Tree Height Transformation based obfuscation :

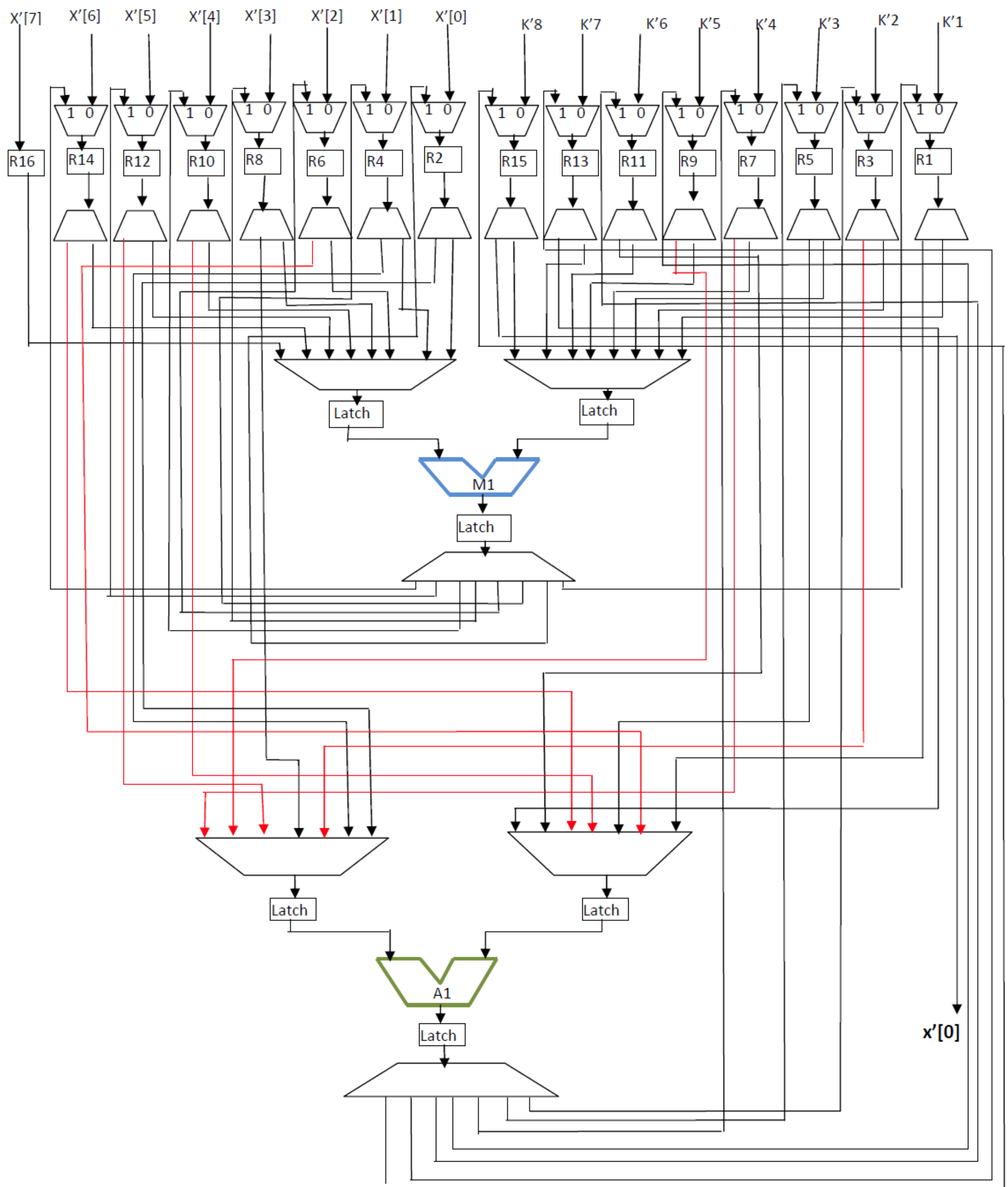


Figure 9

4)

5) FIR Filter: Obfuscation

DFG of FIR after High Level Transformation (Tree Height Transformation):

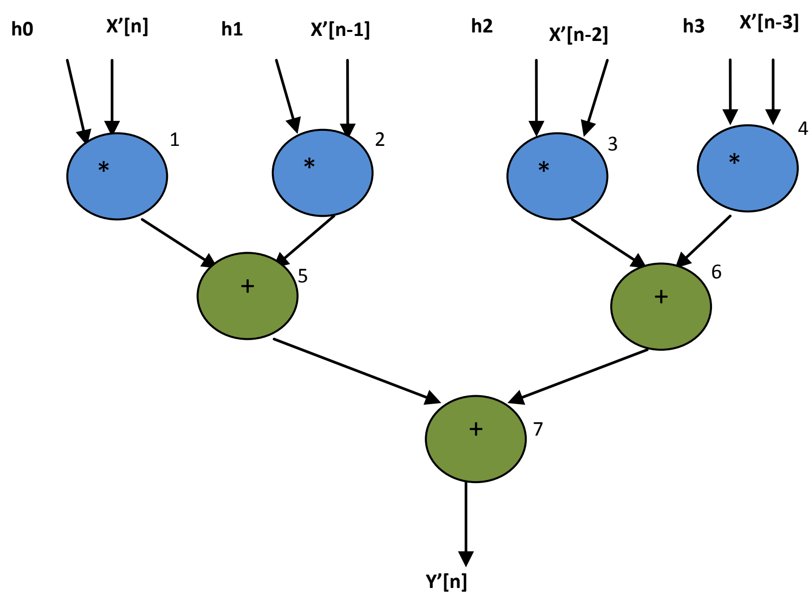


Figure 8.20

Scheduled DFG of FIR based on Resource constraint (1M,1A) after High Level Transformation (Tree Height Transformation):

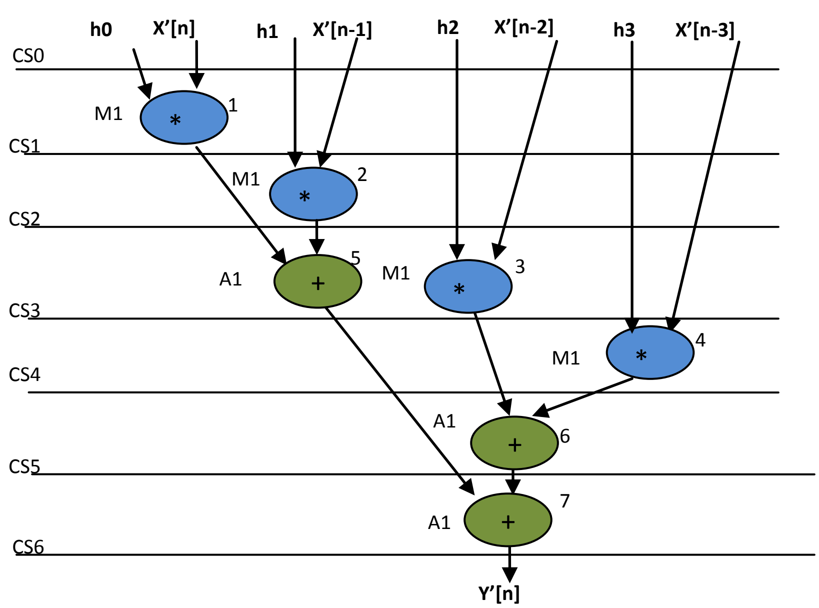


Fig. 8.21

FIR: Y'[n]=h0*X'[n]+ h1*X'[n-1]+ h2*X'[n-2]+ h3*X'[n-3].
Datapath after High level Transformation based obfuscation:

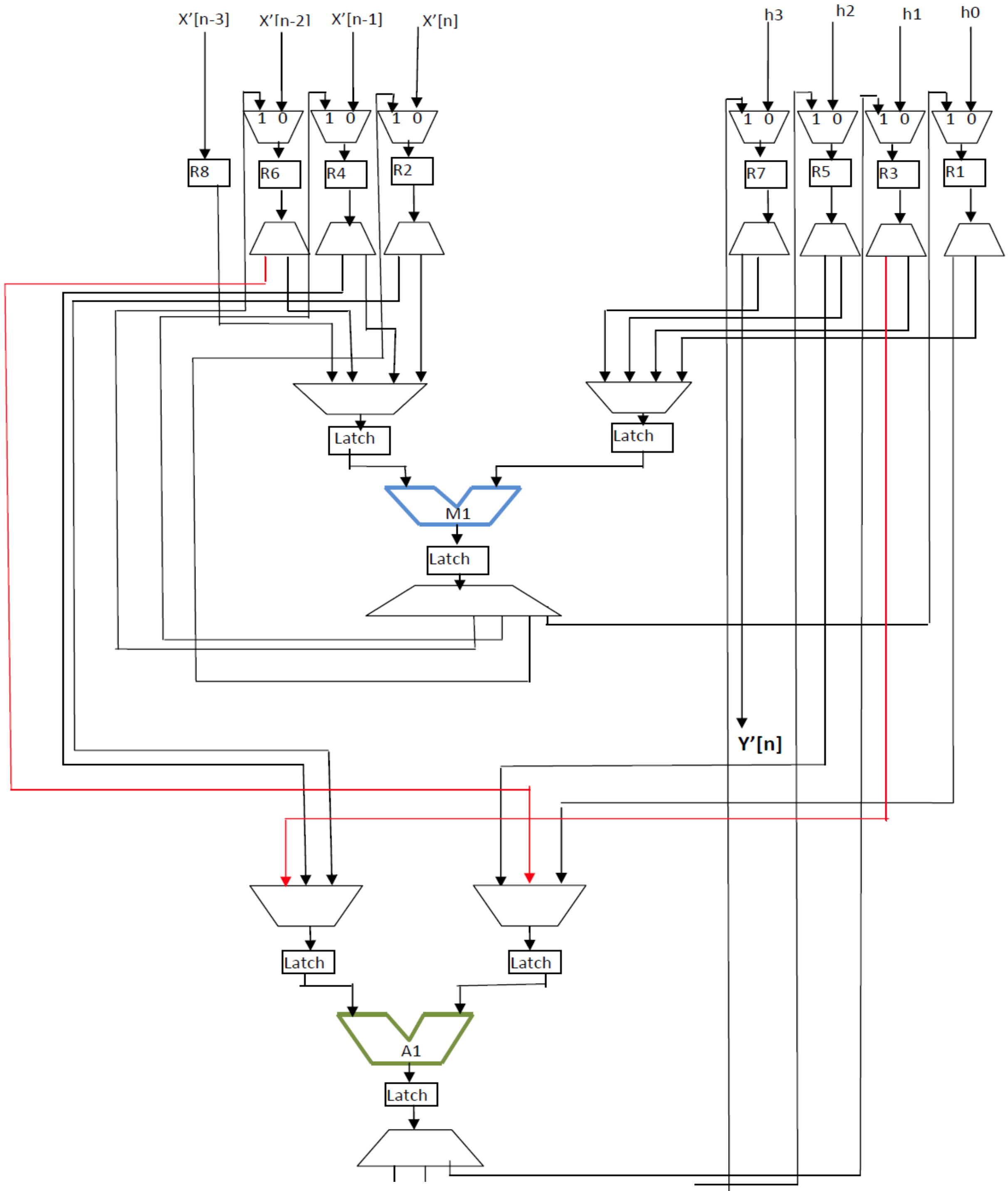


Figure 12

6) IIR Filter: Obfuscation

IIR equation:

Y[n]=b0*X[n]+ b1*X[n-1]+ b2*X[n-2]+b3*X[n-3]+(- a1)*Y[n-1]+(- a2)*Y[n-2]+(-a3)*Y[n-3].

DFG of IIR after High Level Transformation (Tree Height Transformation):

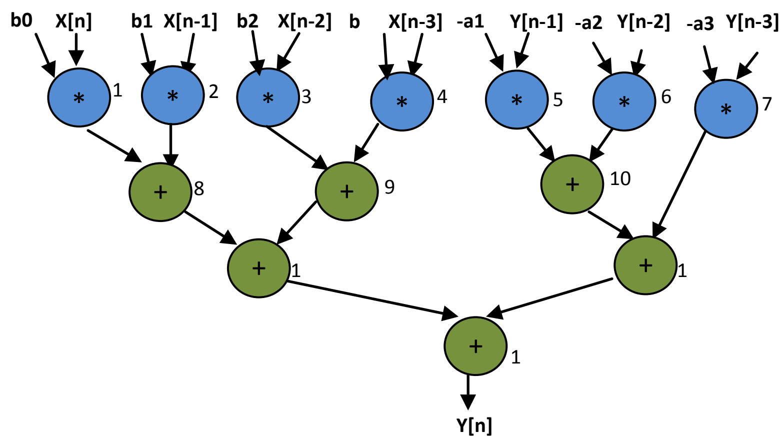


Fig. 13

Scheduled DFG of IIR after High Level Transformation (Tree Height Transformation) Based on Resource constraints (1A, 1M)

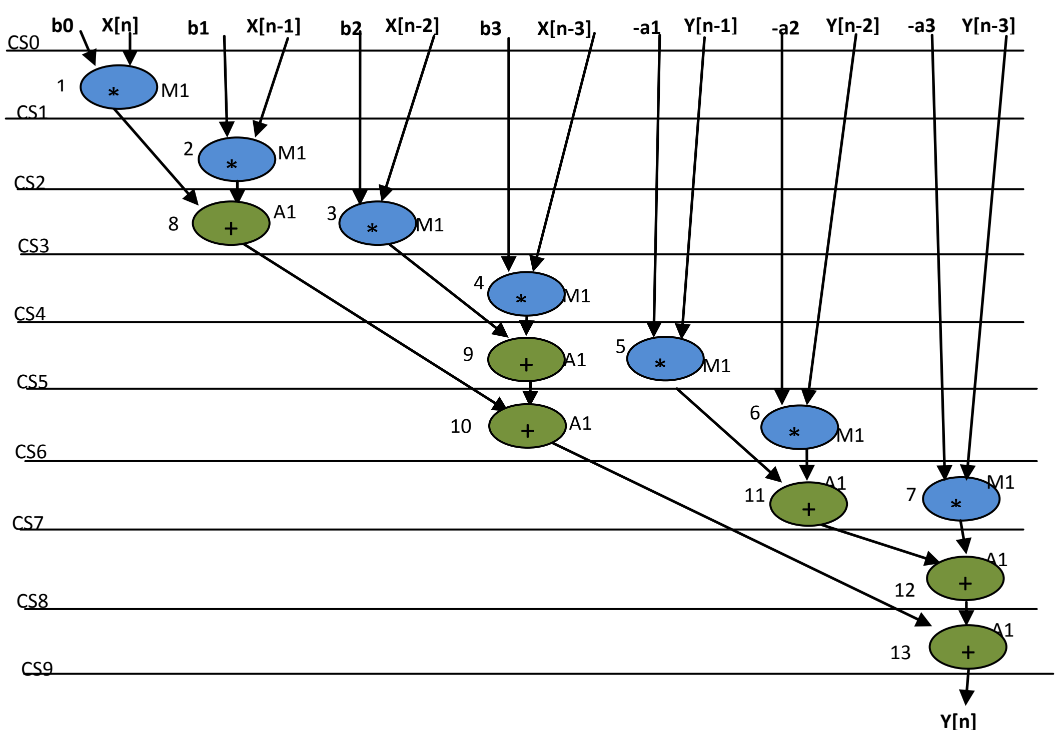


Fig.. 8.24

IIR: Y[n]=b0*X[n]+ b1*X[n-1]+ b2*X[n-2]+b3*X[n-3] +(- a1)*Y[n-1]+(- a2)*Y[n-2]+(- a3)*Y[n-3].

Datapath after High Level Transformation based obfuscation:

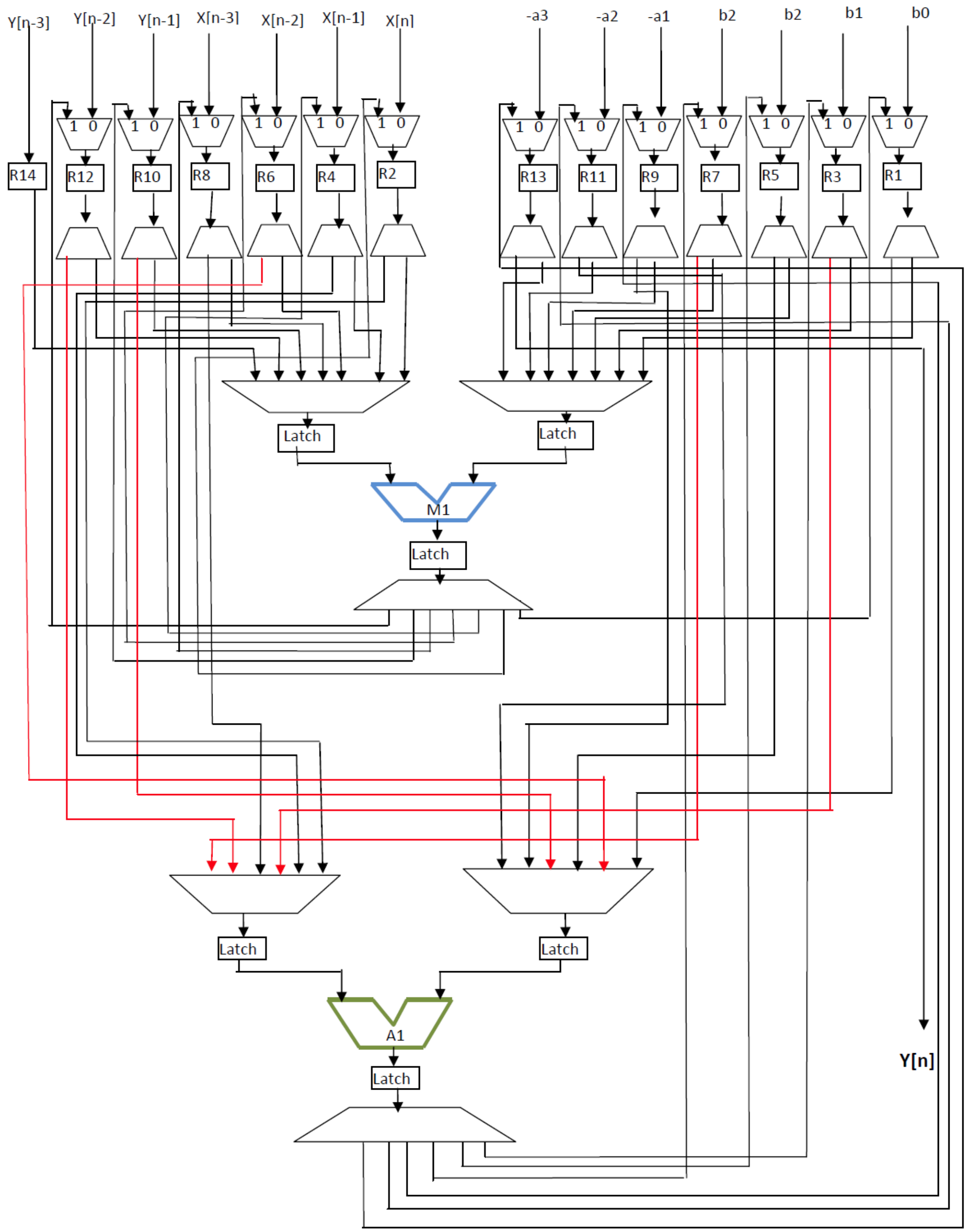


Figure 15

7) 4-point IDCT: Obfuscation

Equations of 4-point IDCT:

x’[0]=c2*X’[0]+ c1*X’[1]+ c2*X’[2]+ c3*X’[3]

x‘[1]=c2*X’[0]+ c3*X’[1]+ (-c2)*X’[2]+ (-c1)*X’[3]

x’[2]=c2*X’[0]+ (-c3)*X’[1]+ (-c2)*X’[2]+ c1*X’[3]

x‘[3]=c2*X’[0]+ (-c1)*X’[1]+ c2*X’[2]+ (-c3)*X’[3]

The generic equation of 4-Point IDCT to compute 1st sample:

x‘[0]=k’1*X’[0]+ k’2*X’[1]+ k’3*X’[2]+ k’4*X’[3]

DFG of 4-point IDCT after Tree Height Transformation based on generic representation to compute 1st sample:

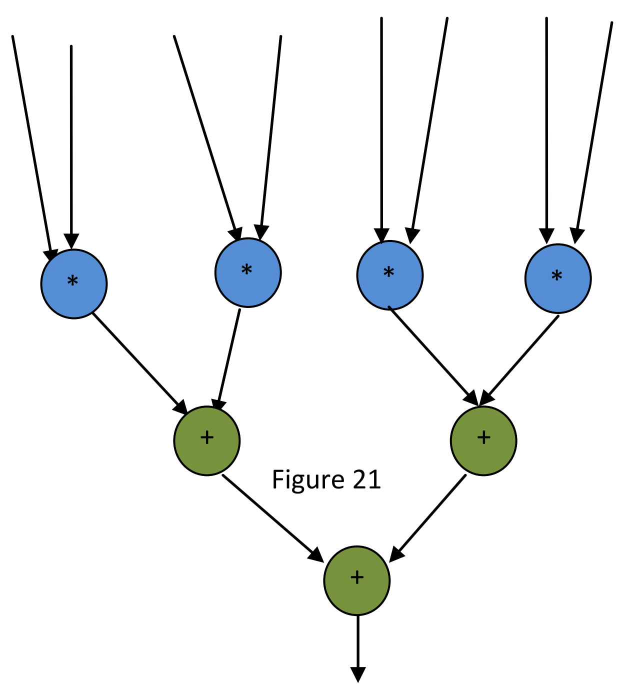


Figure 8. 26

Scheduled DFG of 4-point IDCT based on Resource constraint (1M,1A) after Tree Height Transformation:

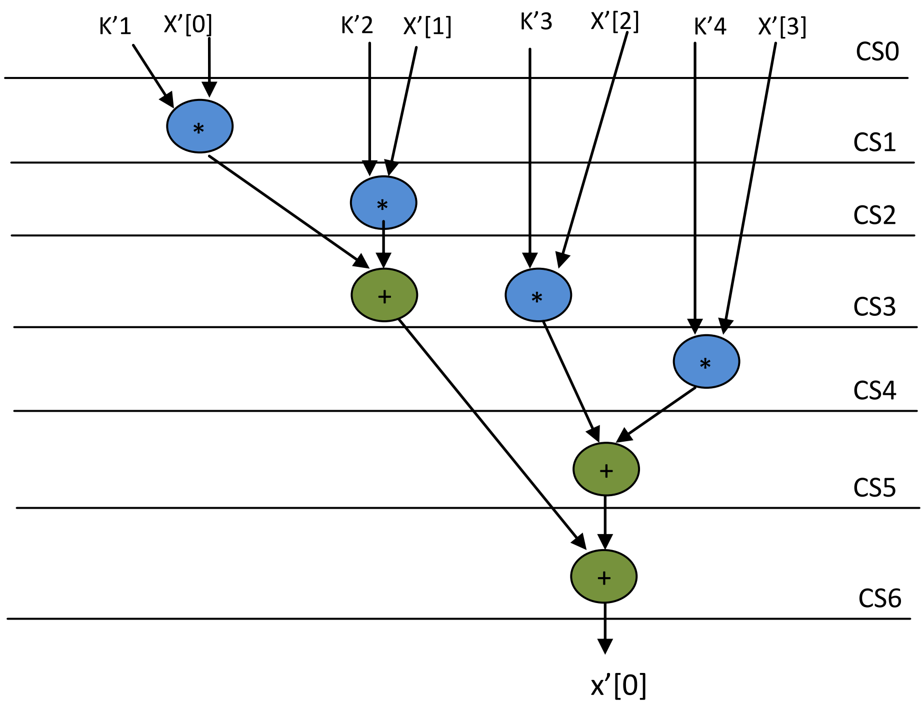
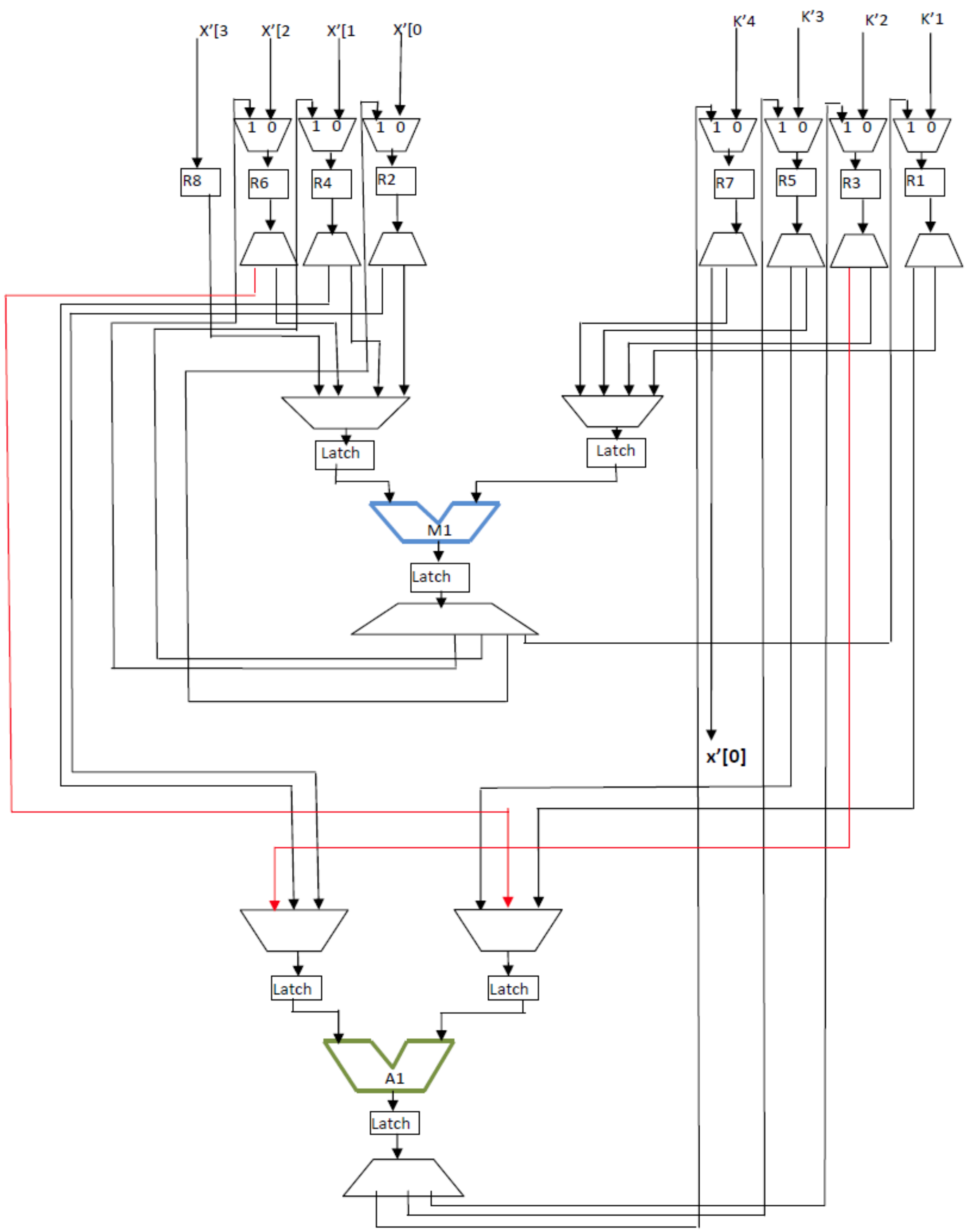


Figure 8.27

Equation of 4-Point IDCT to compute 1st sample: x'[0]=k'1*X'[0]+ k'2*X'[1]+ k'3*X'[2]+ k'4*X'[3]

Datapath after Tree Height transformation based obfuscation:



Appendix Part 2: Non-Obfuscated (Baseline) DSP circuits

DSP Applications – Data Flow Graphs, Scheduling and Datapath Circuits

[A] 4-point DCT

The generic equation of 4-Point DCT to compute 1st sample:

X[0]=k1*x[0]+ k2*x[1]+ k3*x[2]+ k4*x[3]

DFG of 4-point DCT based on generic representation to compute 1st sample:

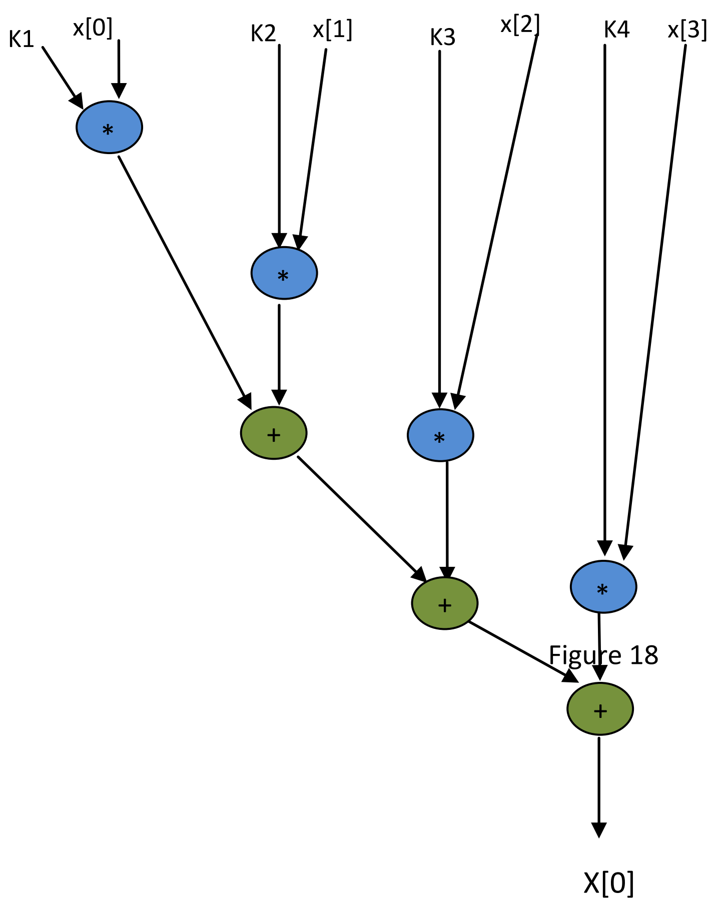


Figure 18

Figure 19

Scheduled DFG of 4-Point DCT Based on Resource constraints (1A, 1M)

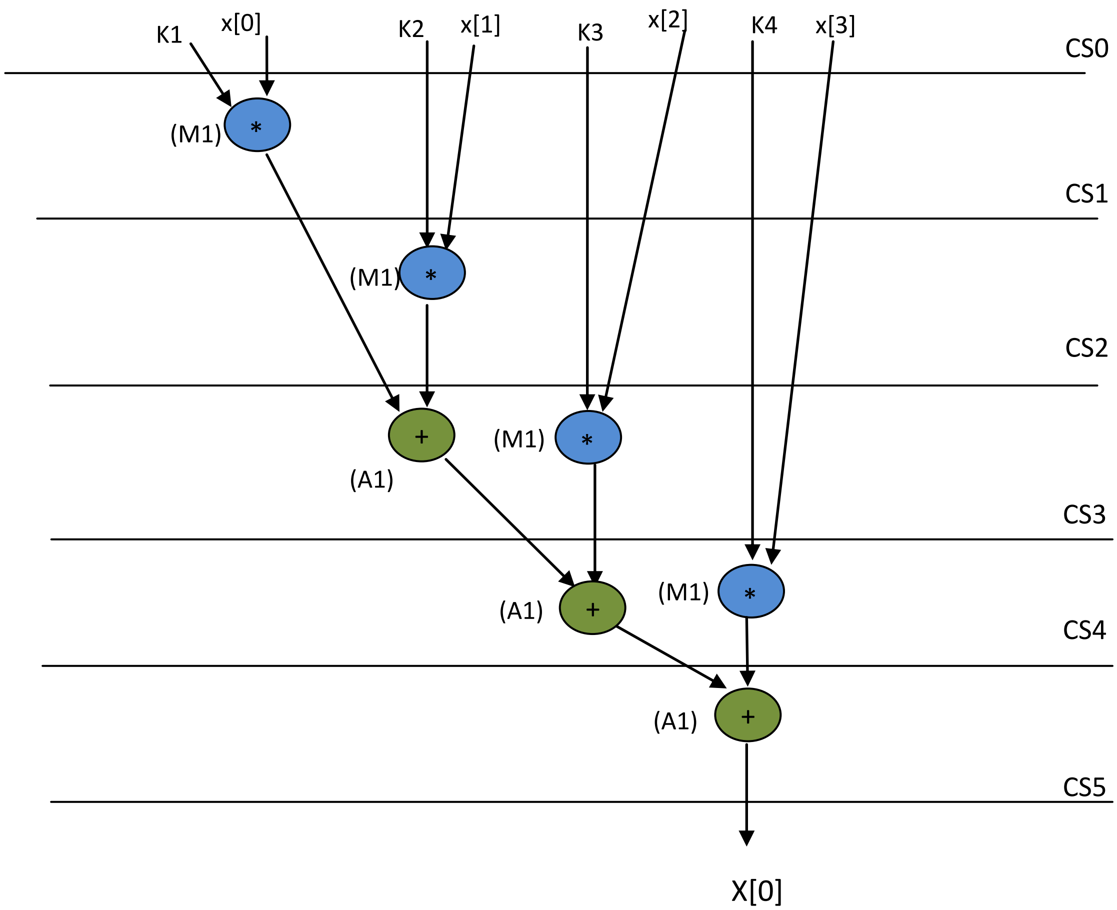
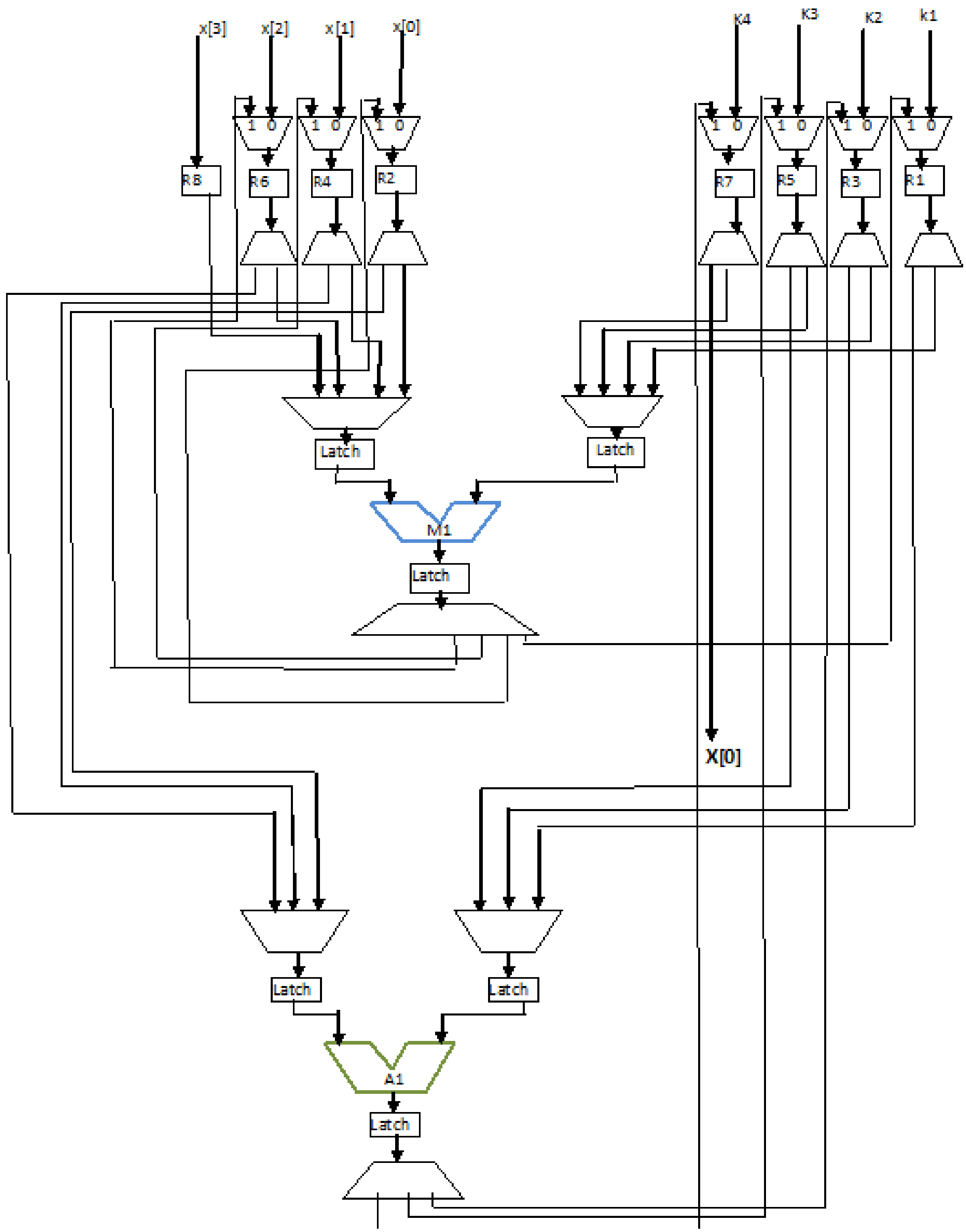


Figure 20

The generic equation of 4-Point DCT to compute 1st sample: X[0]=k1*x[0]+ k2*x[1]+ k3*x[2]+ k4*x[3]. The respective datapath is shown below.



[B] 4-point IDCT

The generic equation of 4-Point IDCT to compute 1st sample:

x‘[0]=k’1*X’[0]+ k’2*X’[1]+ k’3*X’[2]+ k’4*X’[3]

DFG of 4-point IDCT based on generic representation to compute 1st sample:

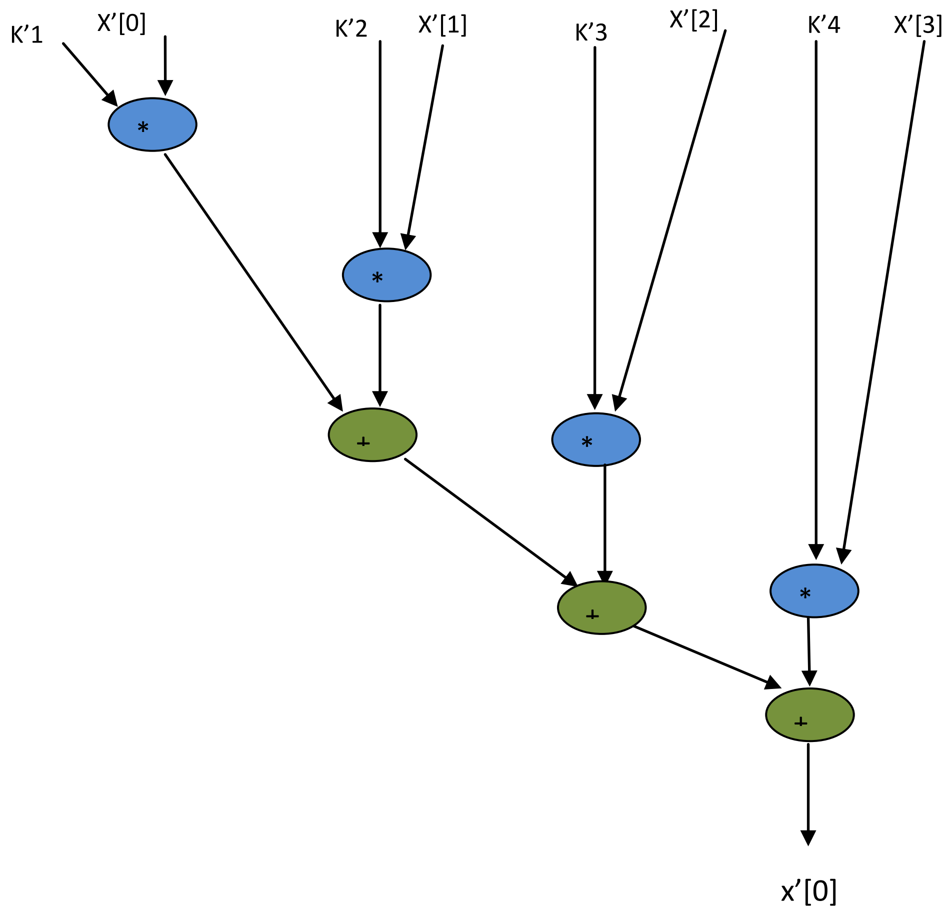


Figure 22

Scheduled DFG of 4-Point IDCT Based on Resource constraints (1A, 1M)

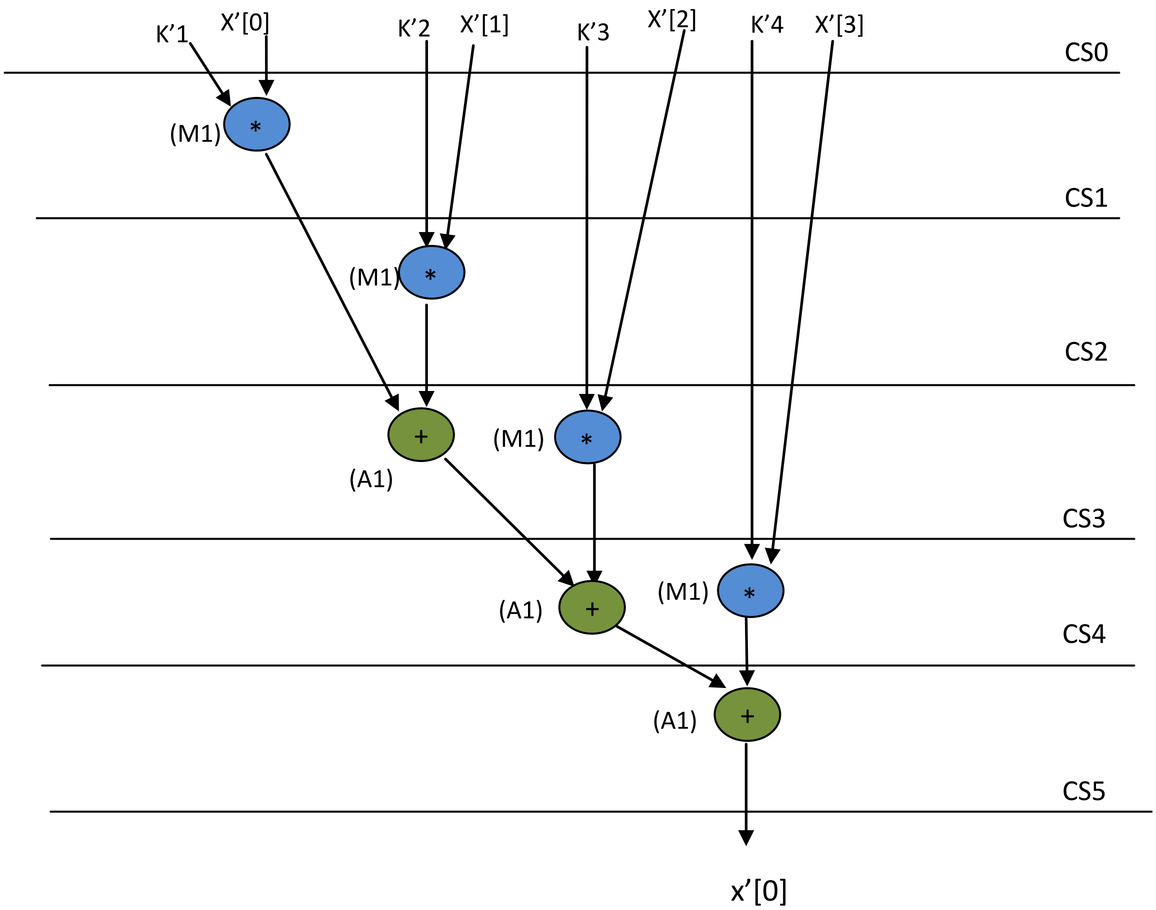


Figure 23

The generic equation of 4-Point IDCT to compute 1st sample: x'[0]=k'1*X'[0]+ k'2*X'[1]+ k'3*X'[2]+ k'4*X'[3]. The respective datapath is shown below.

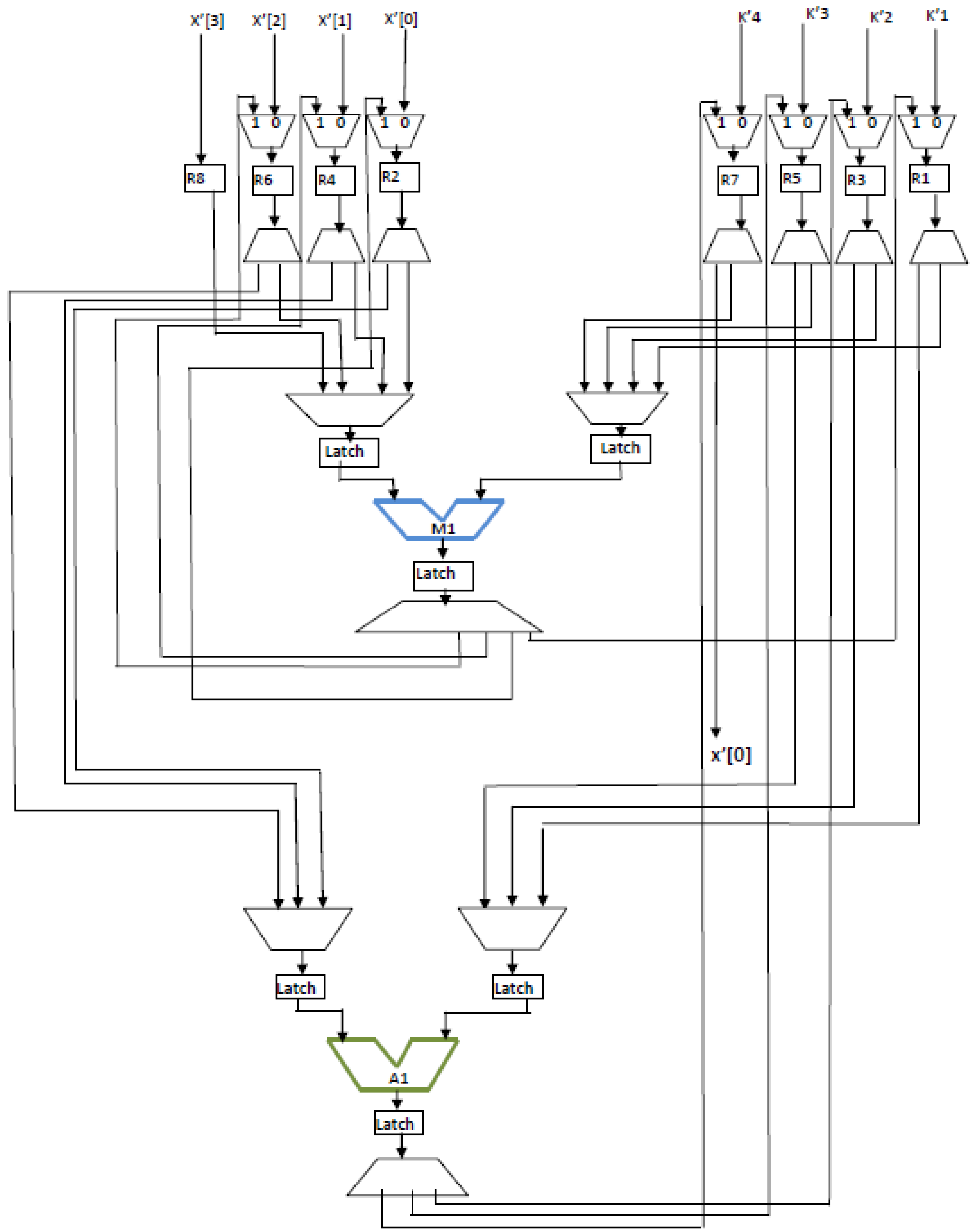


Figure 24

[C] 4-point DIT-FFT

Equations of 4-point DIT-FFT:

X'[0]=H'[0]* W_4^0+G'[0]

X'[1]=H'[1]* W_4^1+G'[1]

X'[2]=H'[0]* W_4^2+G'[0]

X'[3]=H'[1]* W_4^3+G'[1]

DFG of 4-point DIT-FFT:

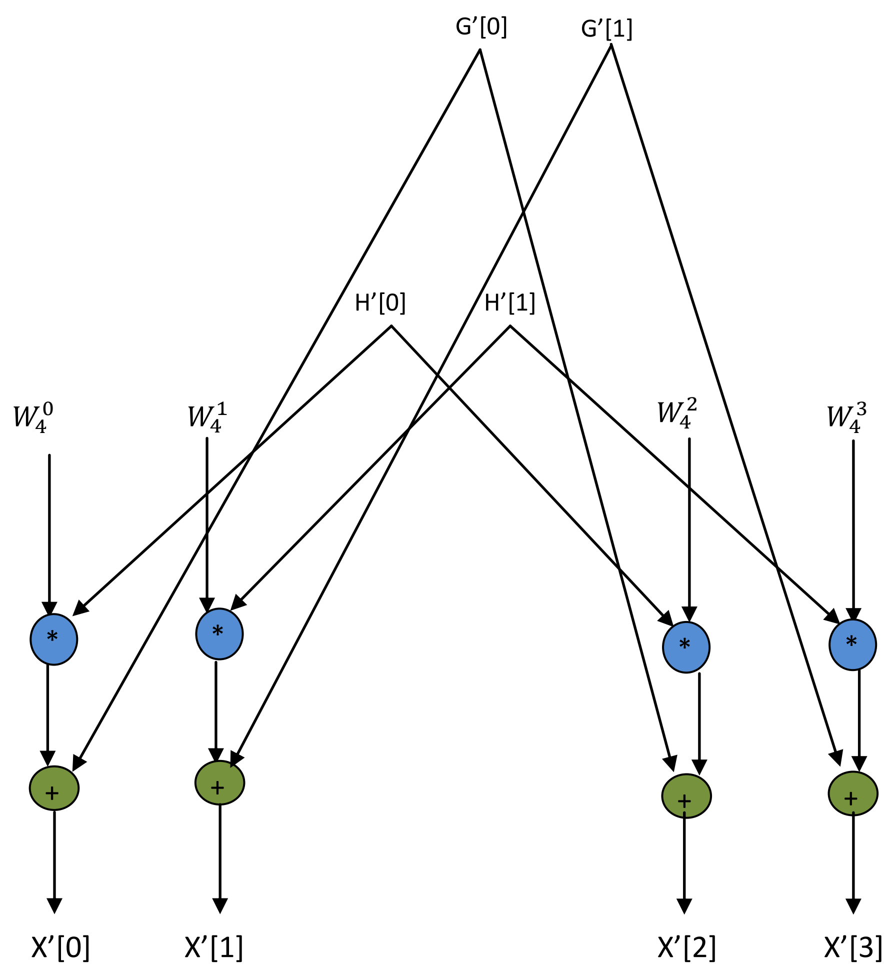


Figure 25

Scheduling of 4-point DIT-FFT:

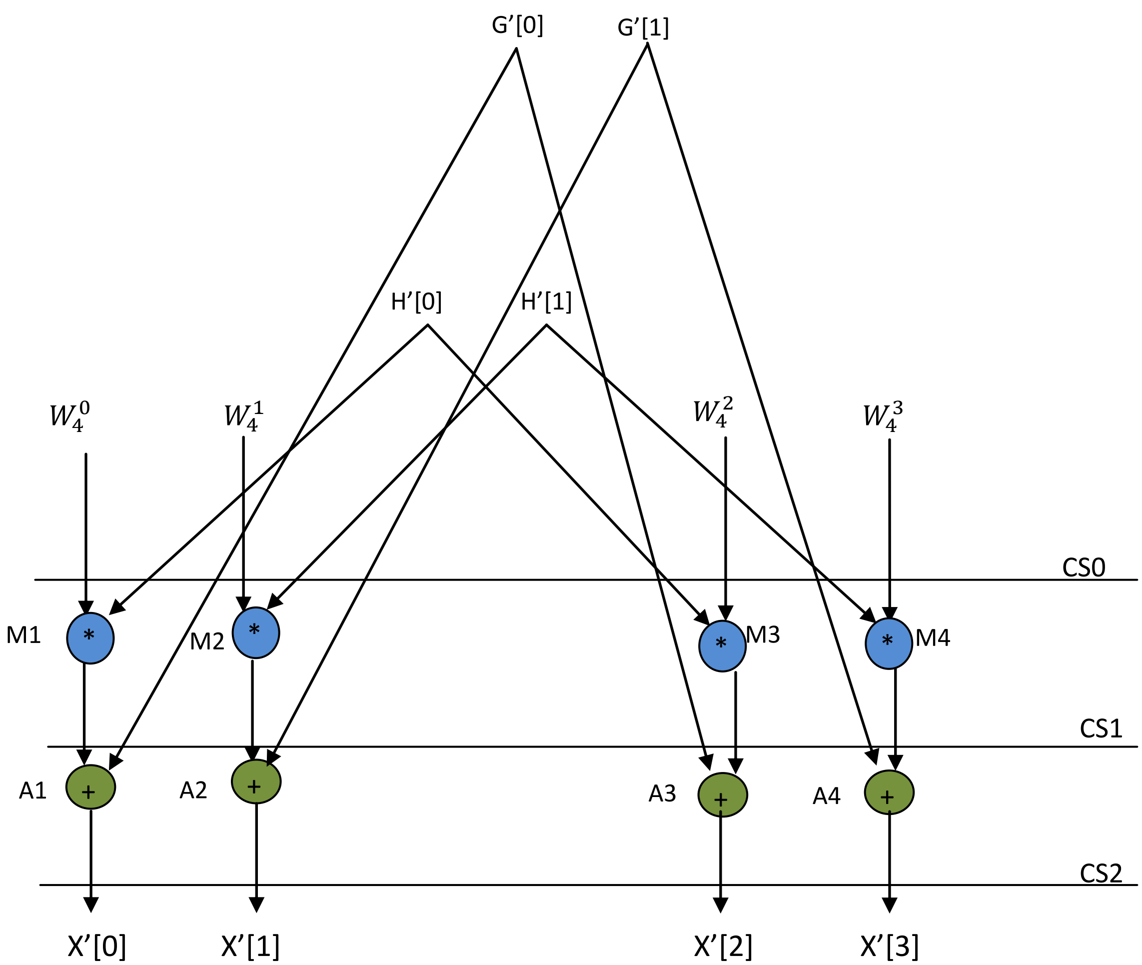


Figure 26

1st and 3rd Equations of 4-point DIT-FFT and corresponding Datapath:

X'[0]=H'[0]* W_4^0+G'[0]

X'[2]=H'[0]* W_4^2+G'[0]

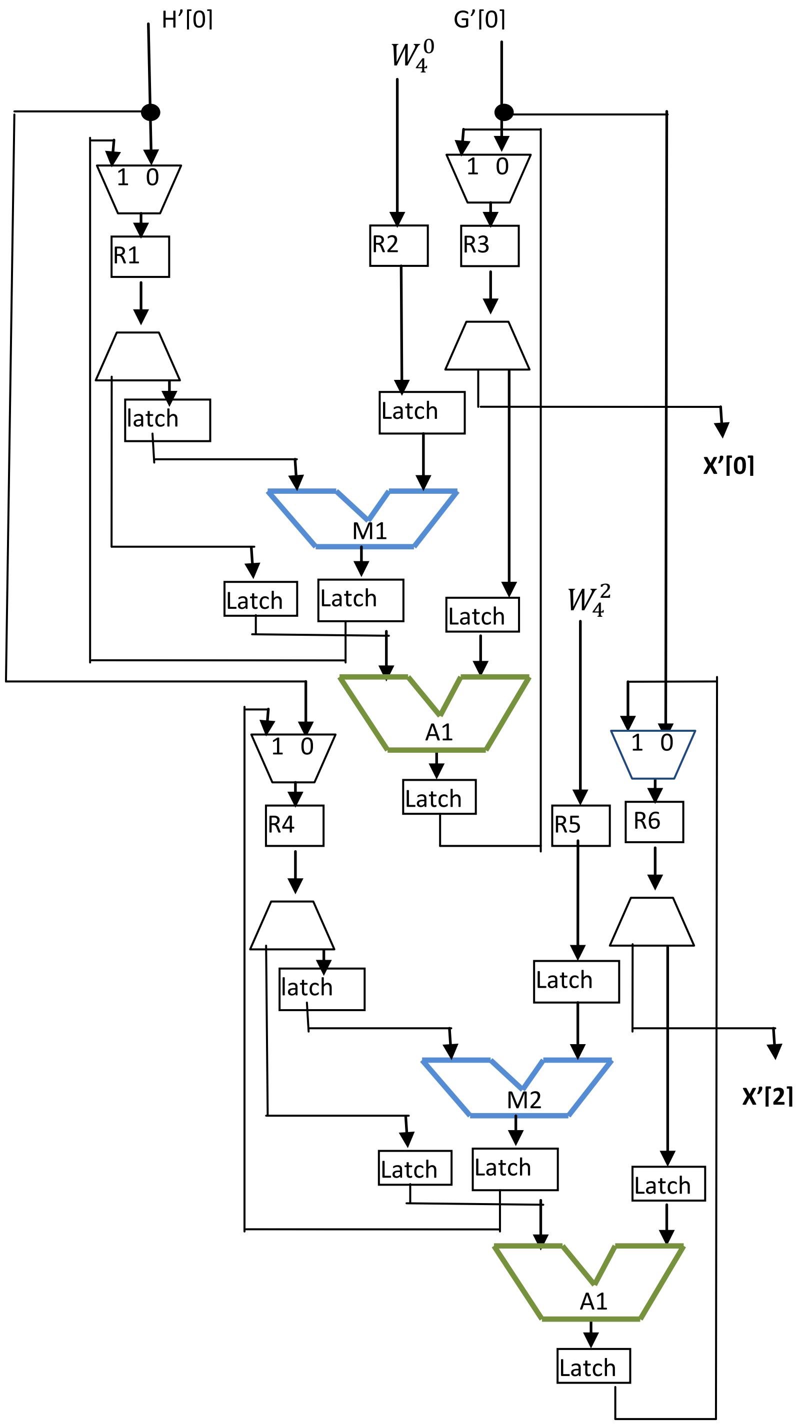


Figure 27

2nd and 4th Equations of 4-point DIT-FFT and corresponding Datapath:

X'[1]=H'[1]* W_4^1+G'[1]

X'[3]=H'[1]* W_4^3+G'[1]

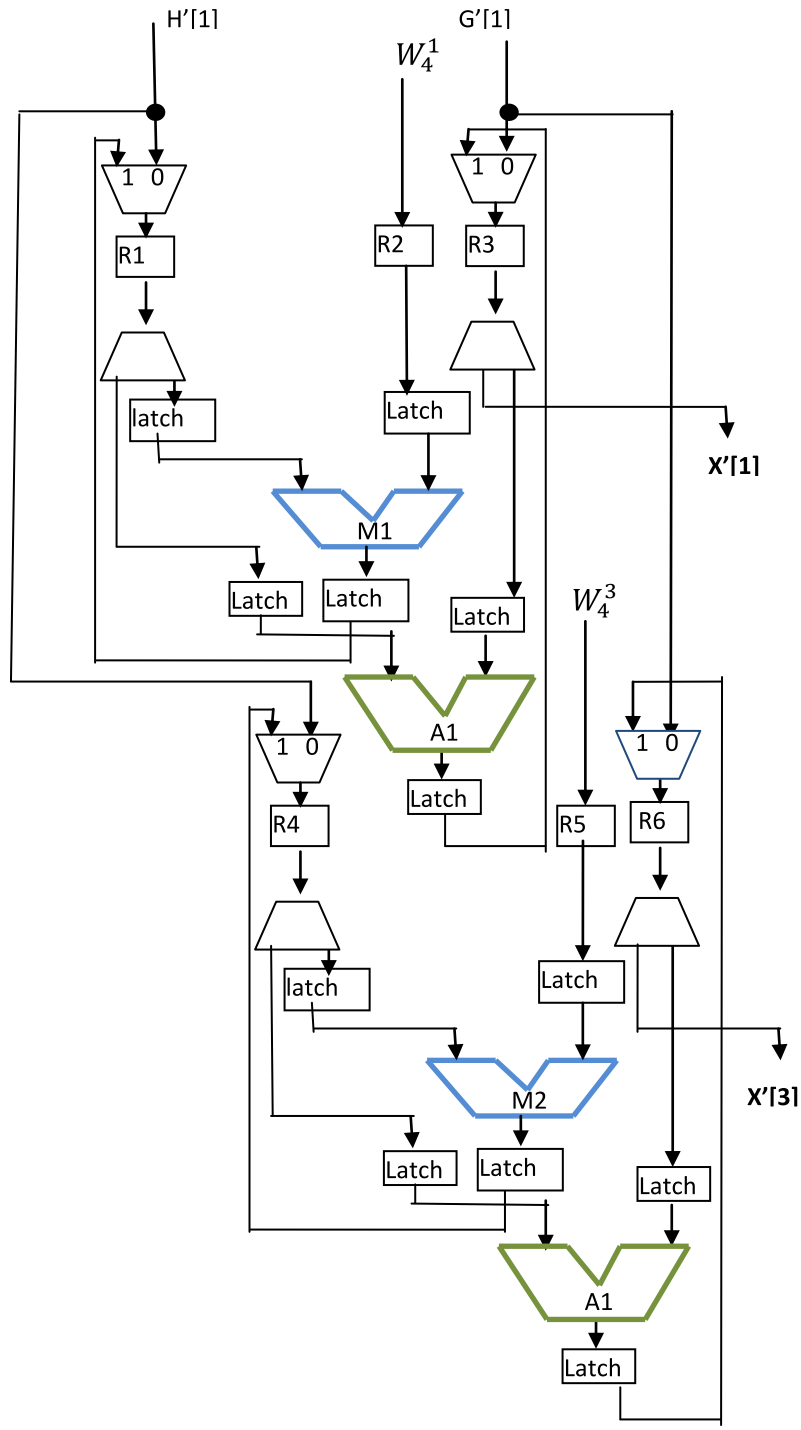


Figure 28

[D] 8-point DIT-FFT

Equations of 8-point DIT-FFT:

X[0]=H[0]* W_8^0+G[0]

X[1]=H[1]* W_8^1+G[1]

X[2]=H[2]* W_8^2+G[2]

X[3]=H[3]* W_8^3+G[3]

X[4]=H[0]* W_8^4+G[0]

X[5]=H[1]* W_8^5+G[1]

X[6]=H[2]* W_8^6+G[2]

X[7]=H[3]* W_8^7+G[3]

DFG of 8-point DIT-FFT:

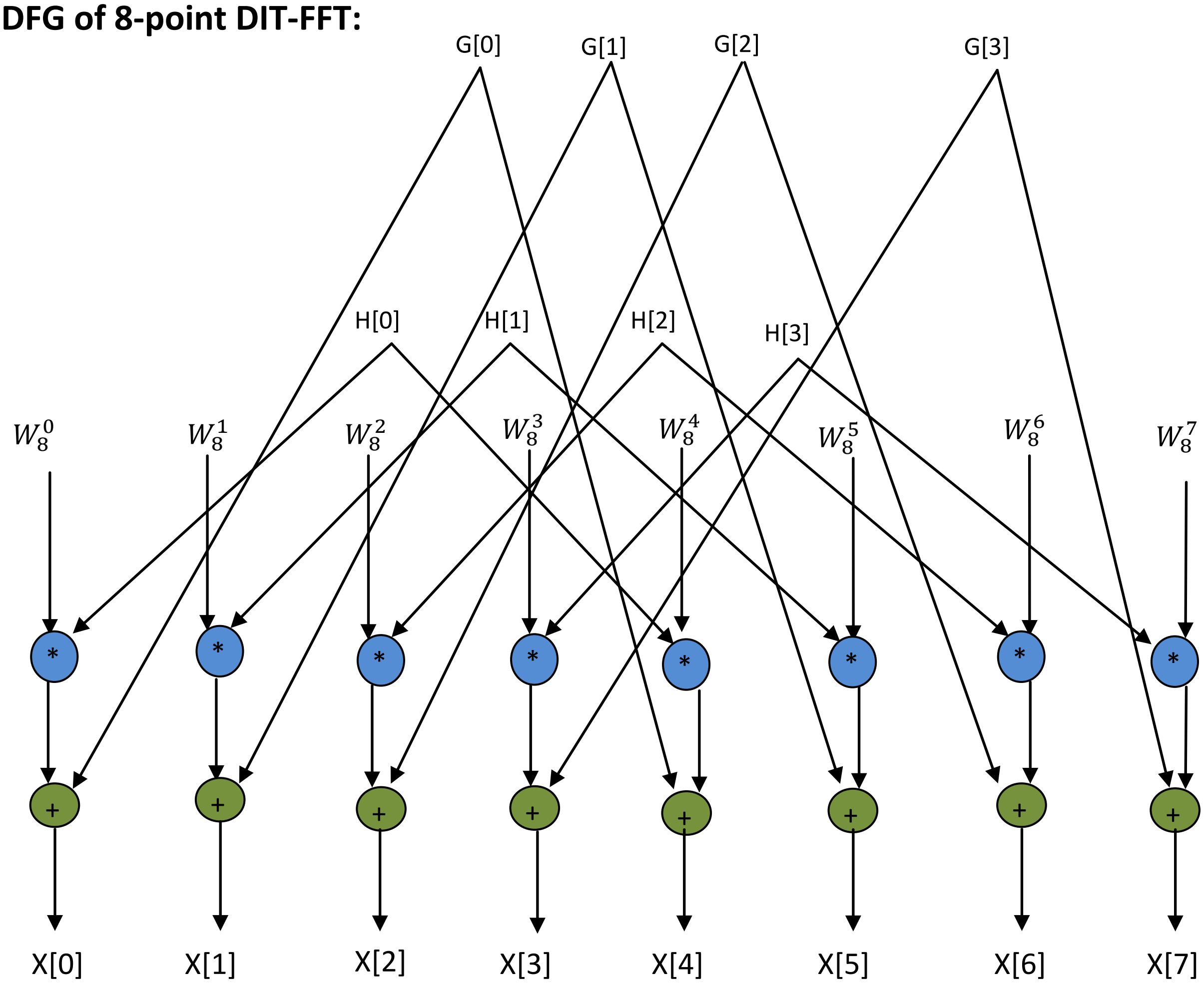


Figure 29

Scheduled DFG of 8-point DIT-FFT:

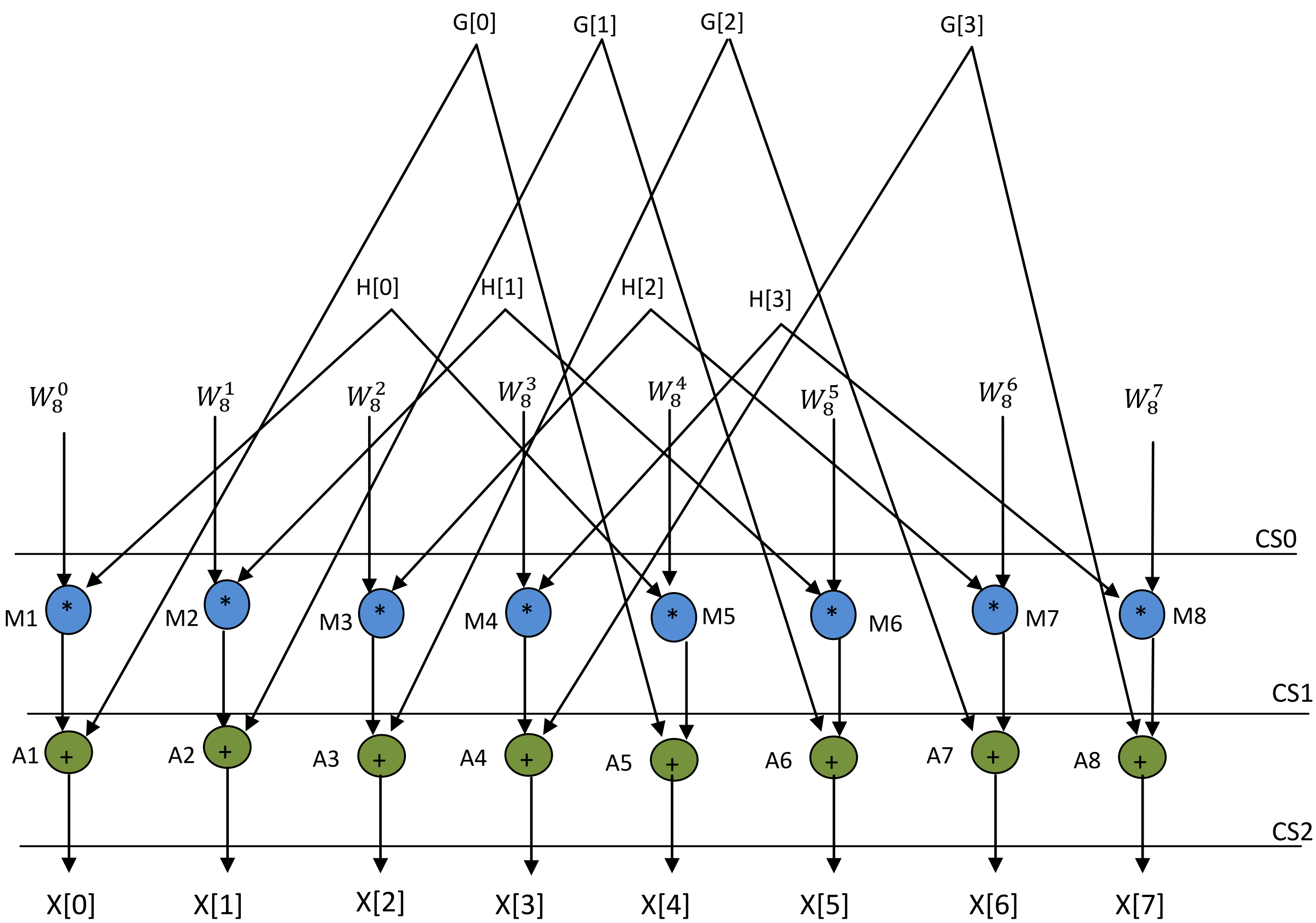


Figure 30

1st and 5th Equations of 8-point DIT-FFT and corresponding Datapath:

$X[0]=H[0]*W_8^0+G[0]$

$X[4]=H[0]*W_8^4+G[0]$

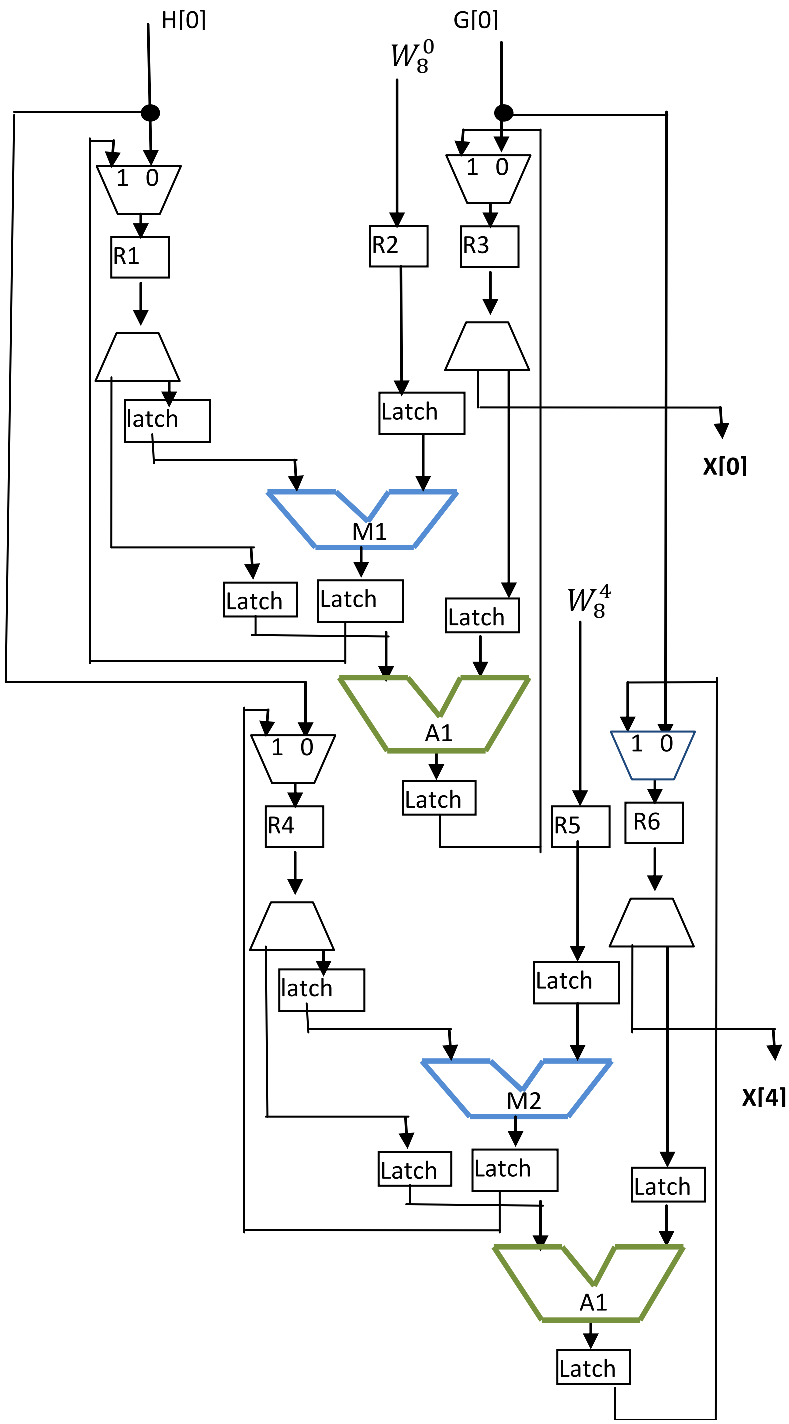


Figure 31

2nd and 6th Equations of 8-point DIT-FFT and corresponding Datapath:

$X[1]=H[1]*W_8^1+G[1]$

$X[5]=H[1]*W_8^5+G[1]$

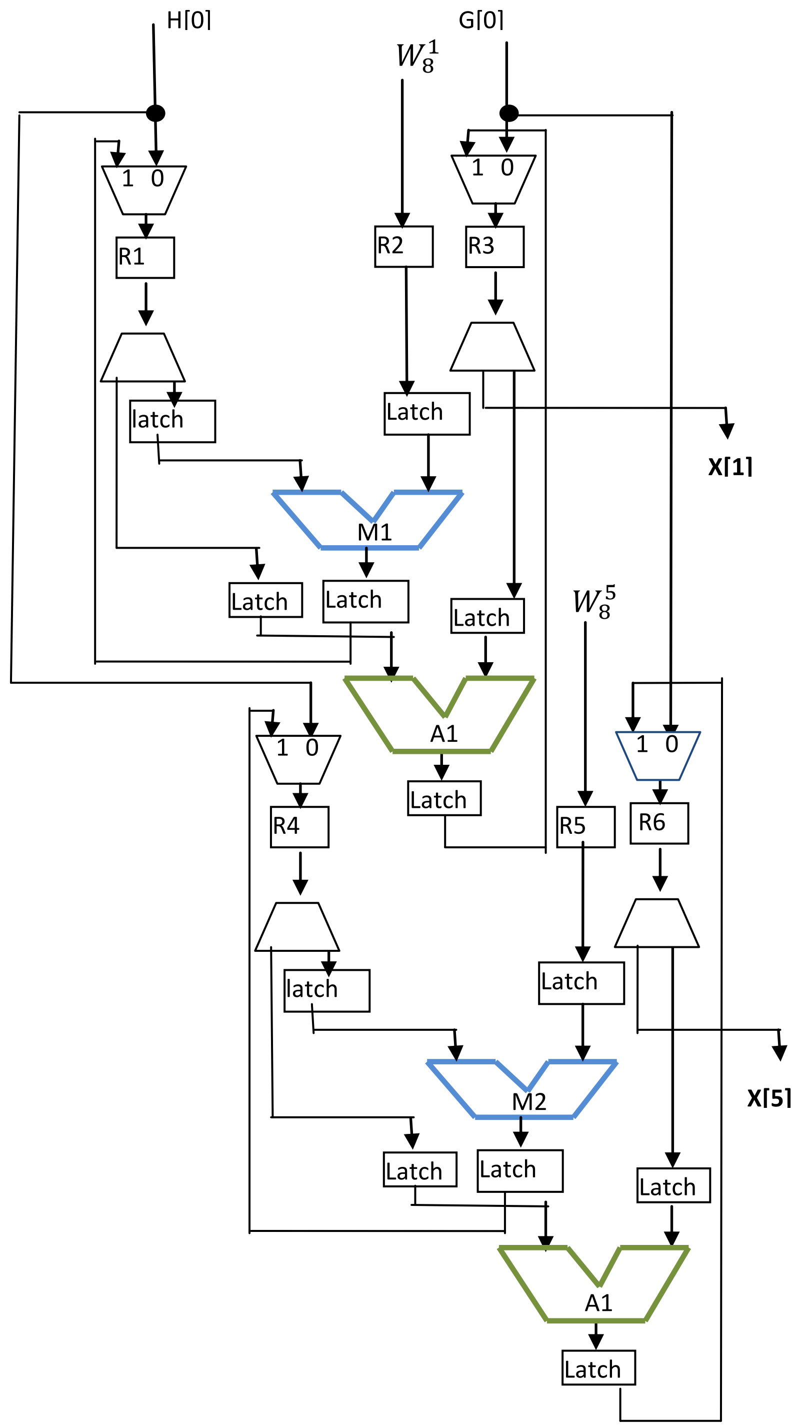


Figure 32

3rd and 7th Equations of 8-point DIT-FFT and corresponding Datapath:

X[2]=H[2]* W_8^2+G[2]

X[6]=H[2]* W_8^6+G[2]

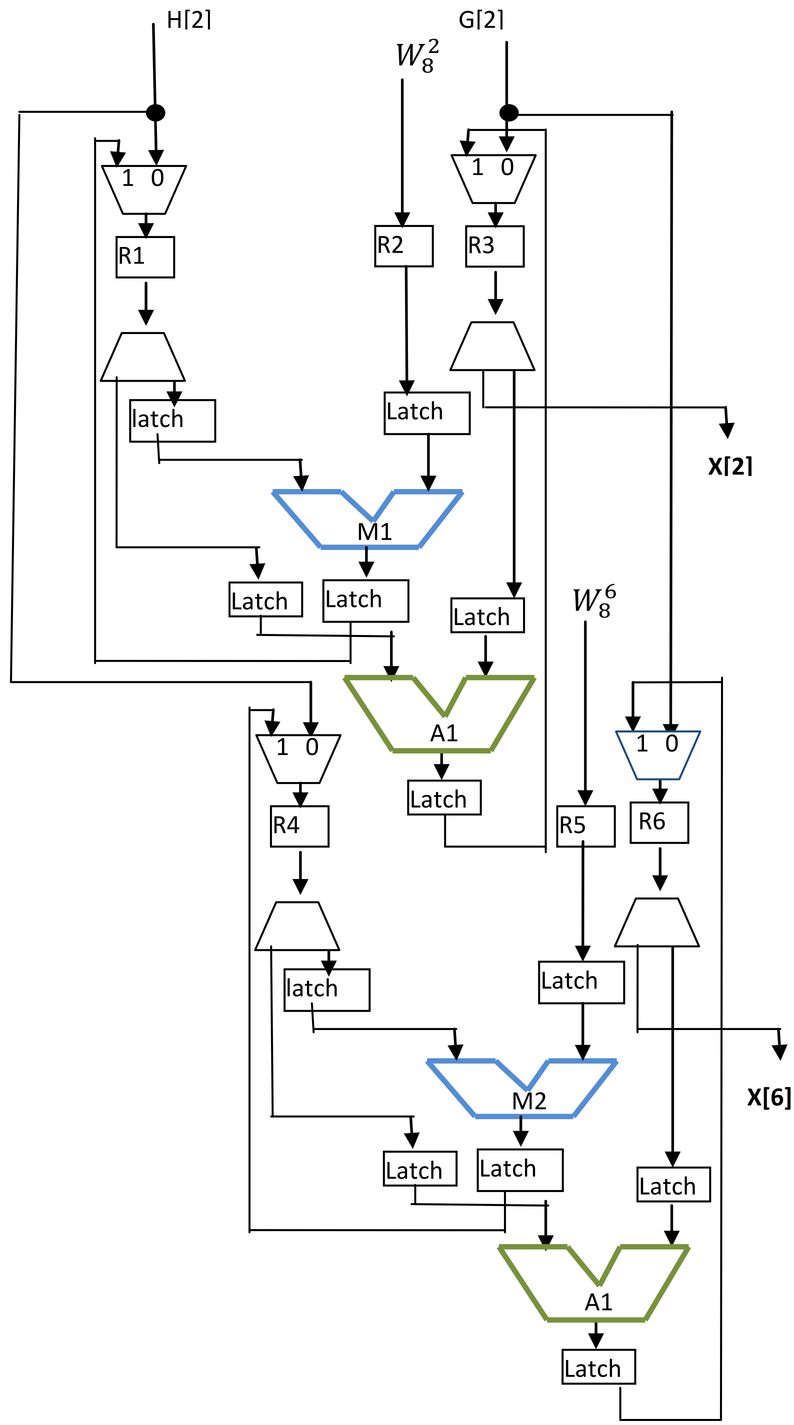


Figure 33

4^{th} and 8^{th} Equations of 8-point DIT-FFT and corresponding Datapath:

X[3]=H[3]* W_8^3+G[3]

X[7]=H[3]* W_8^7+G[3]

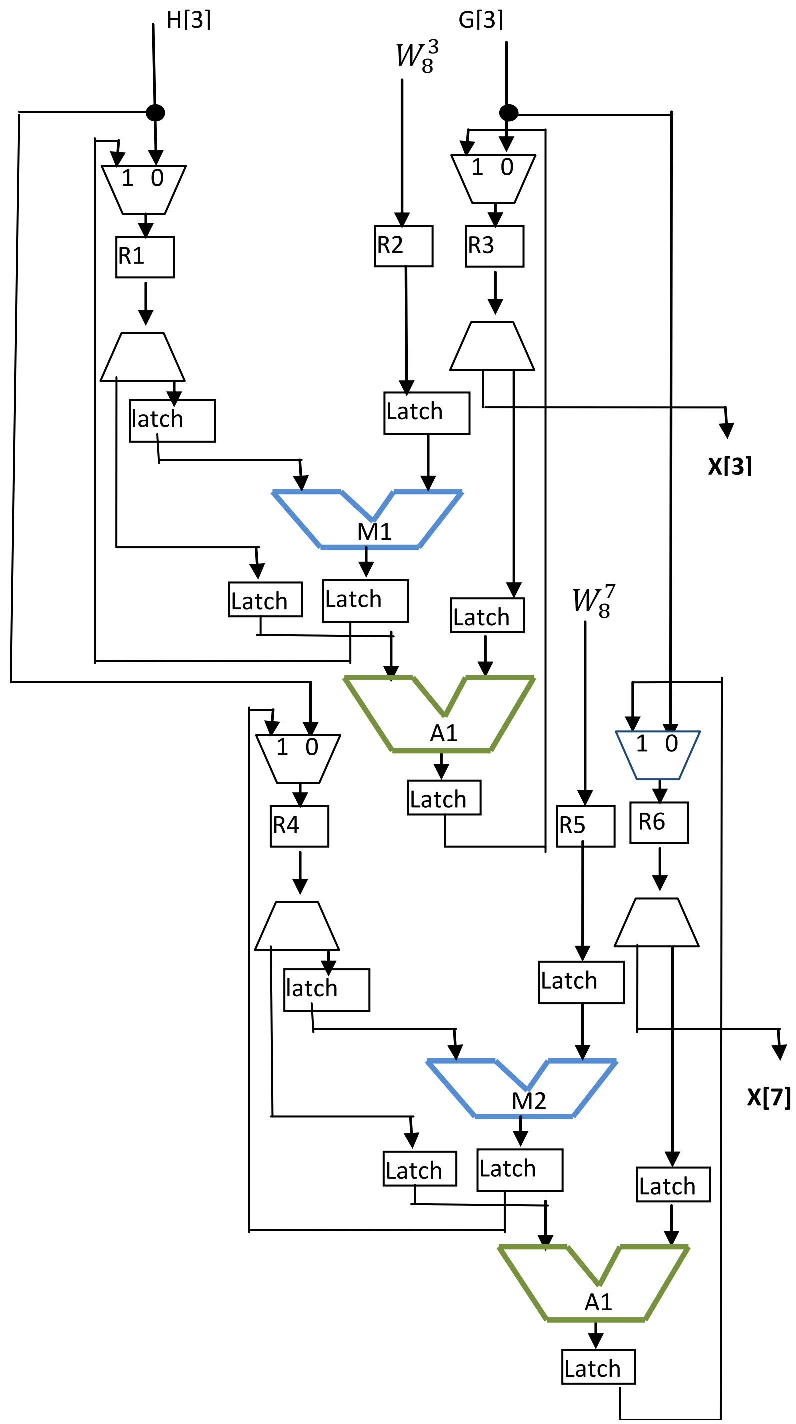


Figure 34

[E] 8-point DCT

Equations of 8-point DCT:

X[0]=c4*x[0]+c4*x[1]+c4*x[2]+c4*x[3]+c4*x[4]+ c4*x[5]+ c4*x[6]+ c4*x[7]

X[1]=c1*x[0]+c3*x[1]+c5*x[2]+c7*x[3]+(-c7)*x[4]+(- c5)*x[5]+ (-c3)*x[6]+ (-c1)*x[7]

X[2]=c2*x[0]+c6*x[1]+(-c6)*x[2]+(-c2)*x[3]+(-c2)*x[4]+(- c6)*x[5]+ c6*x[6]+ c2*x[7]

X[3]=c3*x[0]+(-c7)*x[1]+(-c1)*x[2]+(-c5)*x[3]+c5*x[4]+ c1*x[5]+ c7*x[6]+ (-c3)*x[7]

X[4]=c4*x[0]+(-c4)*x[1]+(-c4)*x[2]+c4*x[3]+c4*x[4]+ (-c4)*x[5]+ (-c4)*x[6]+ c4*x[7]

X[5]=c5*x[0]+(-c1)*x[1]+c7*x[2]+c3*x[3]+(-c3)*x[4]+ (-c7)*x[5]+ c1*x[6]+ (-c5)*x[7]

X[6]=c6*x[0]+(-c2)*x[1]+c2*x[2]+(-c6)*x[3]+(-c6)*x[4]+ c2*x[5]+ (-c2)*x[6]+ c6*x[7]

X[7]=c7*x[0]+(-c5)*x[1]+c3*x[2]+(-c1)*x[3]+c1*x[4]+ (-c3)*x[5]+ c5*x[6]+ (-c7)*x[7]

Generic representation of 8-Point DCT to compute 1st sample:

X[0]=k1*x[0]+ k2*x[1]+ k3*x[2]+ k4*x[3]+ k5*x[4]+ k6*x[5]+ k7*x[6]+ k8*x[7]

DFG of Generic representation of 8-Point DCT to compute 1st sample:

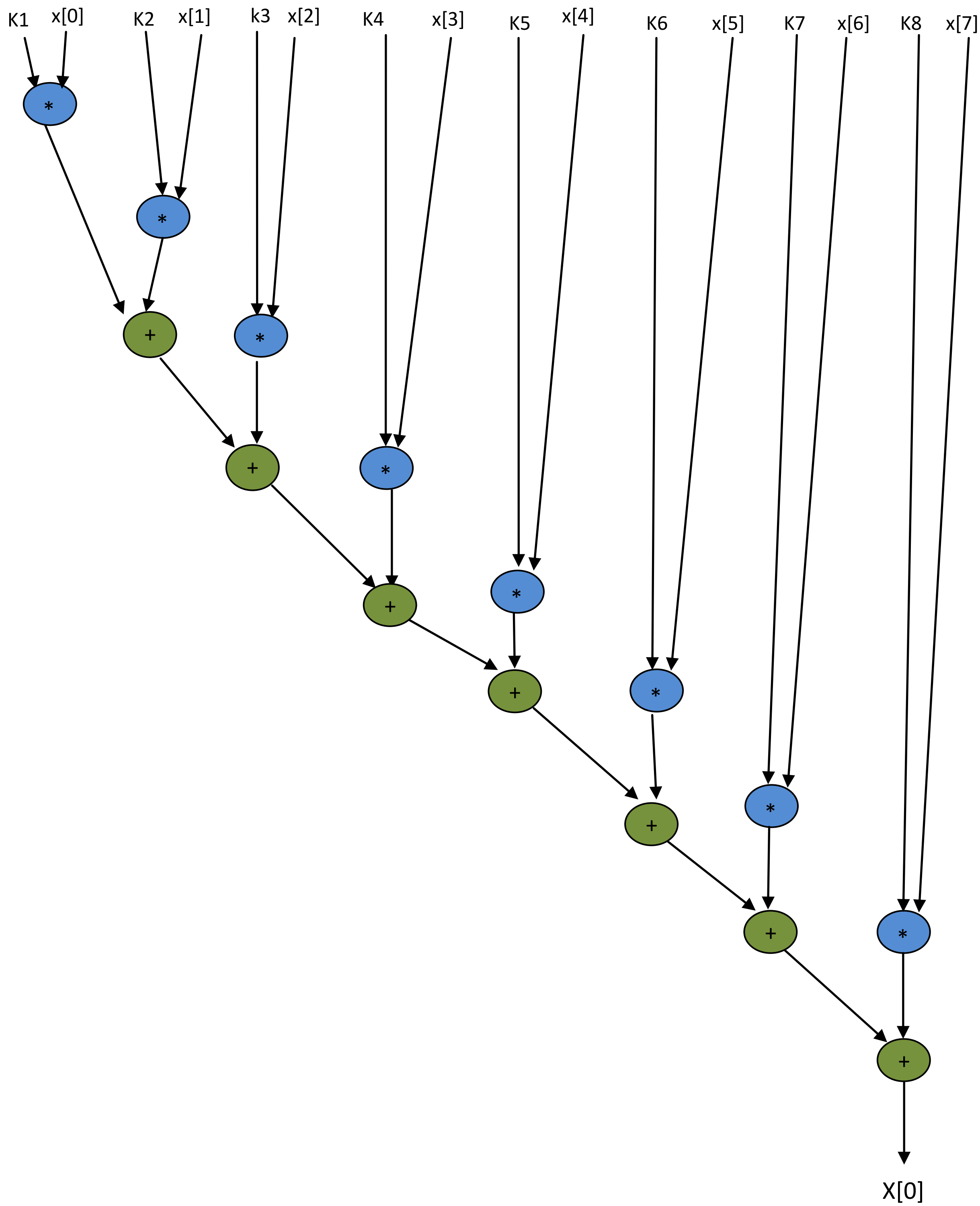


Figure 35

Scheduled DFG of 8-point DCT based on Resource constraints (1A, 1M).

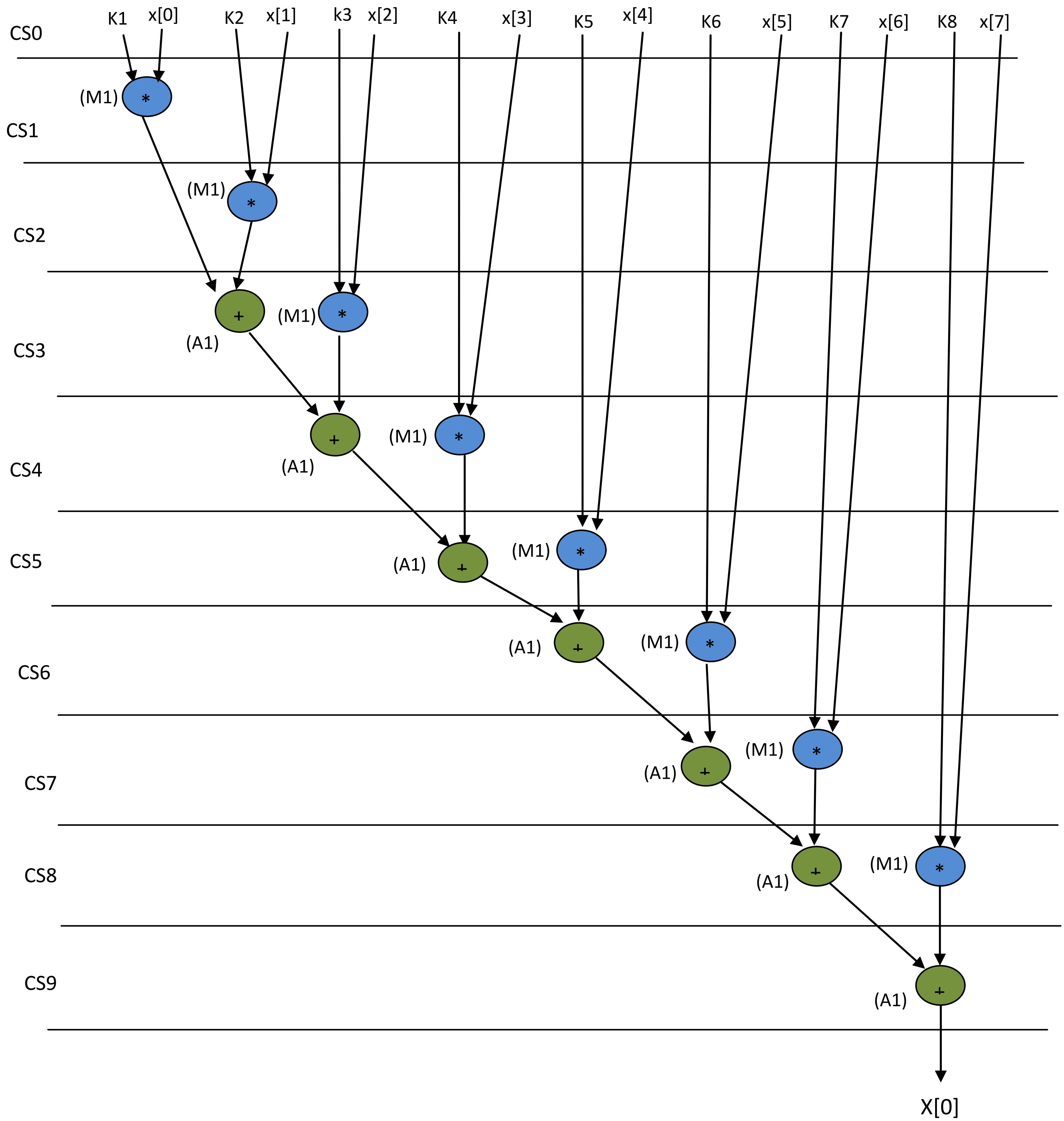


Figure 36

8-Point DCT: X[0]=k1*x[0]+ k2*x[1]+ k3*x[2]+ k4*x[3]+ k5*x[4]+ k6*x[5]+ k7*x[6]+ k8*x[7]. Th respective datapath is shown below.

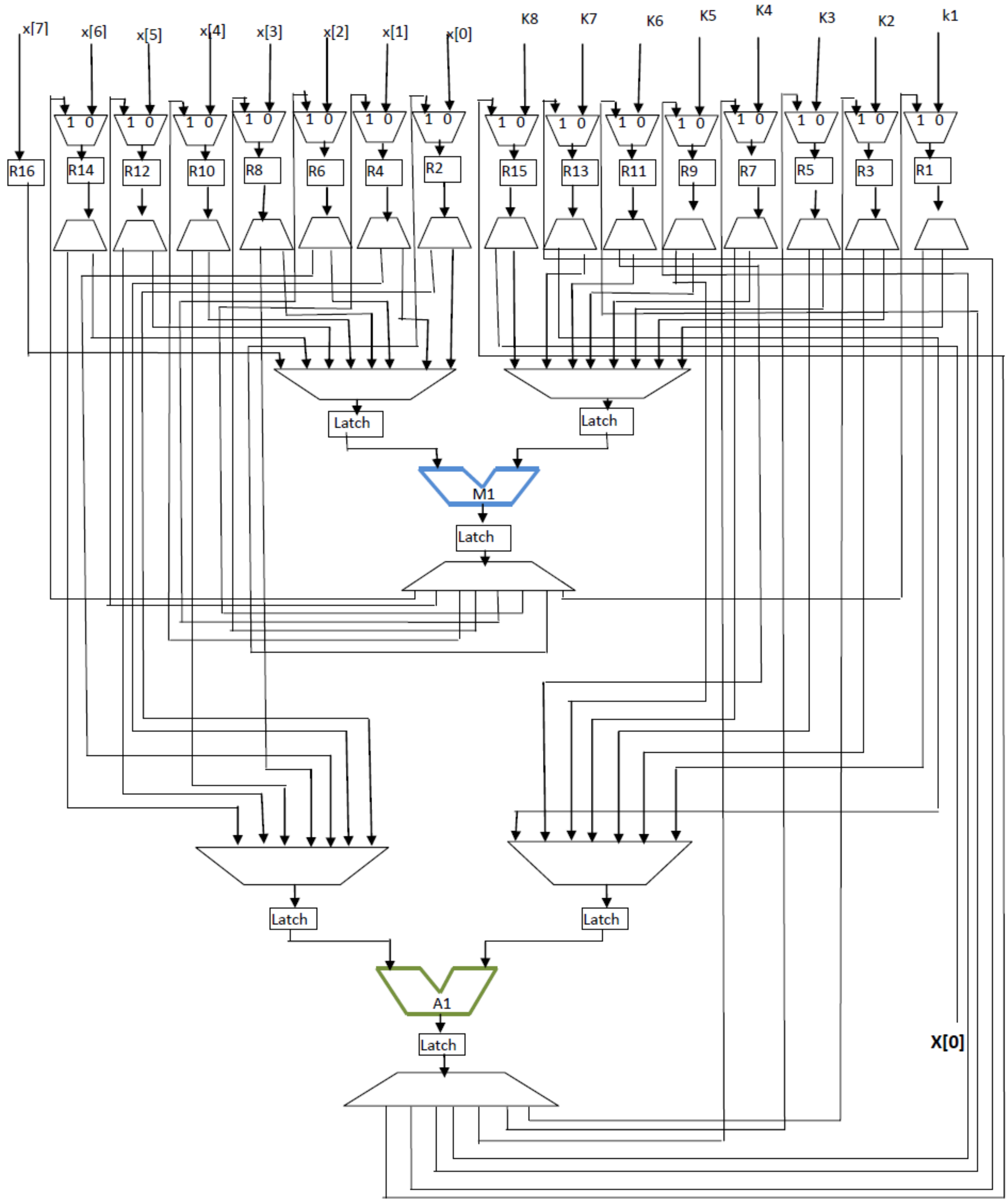


Figure 37

[F] 8-point IDCT

Equations of 8-point IDCT:

$$x'[0]=c4*X'[0]+c1*X'[1]+c2*X'[2]+c3*X'[3]+c4*X'[4]+ c5*X'[5]+ c6*X'[6]+ c7*X'[7]$$

$$x'[1]=c4*X'[0]+c3*X''[1]+c6*X'[2]+(-c7)*X'[3]+(-c4)*X'[4]+(- c1)*X'[5]+ (-c2)*X'[6]+ (-c5)*X'[7]$$

$$x'[2]=c4*X'[0]+c5*X'[1]+(-c6)*X'[2]+(-c1)*X'[3]+(-C4)*X'[4]+ c7*X'[5]+ c2*X'[6]+ c3*X'[7]$$

$$x'[3]=c4*X'[0]+c7*X'[1]+(-c2)*X'[2]+(-c5)*X'[3]+c4*X'[4]+ c3*X'[5]+ (-c6)*X'[6]+ (-c1)*X'[7]$$

$$x'[4]=c4*X'[0]+(-c7)*X'[1]+(-c2)*X'[2]+c5*X'[3]+c4*X'[4]+ (-c3)*X'[5]+ (-c6)*X'[6]+ c1*X'[7]$$

$$x'[5]=c4*X'[0]+(-c5)*X'[1]+(-c6)*X'[2]+c1*X'[3]+(-c4)*X'[4]+ (-c7)*X'[5]+ c2*X'[6]+ (-c3)*X'[7]$$

$$x'[6]=c4*X'[0]+(-c3)*X'[1]+c6*X'[2]+c7*X'[3]+(-c4)*X'[4]+ c1*X'[5]+ (-c2)*X'[6]+ c5*X'[7]$$

$$x'[7]=c4*X'[0]+(-c1)*X'[1]+c2*X'[2]+(-c3)*X'[3]+c4*X'[4]+ (-c5)*X'[5]+ c6*X'[6]+ (-c7)*X'[7]$$

Generic representation of 8-Point IDCT to compute 1st sample:

$$x'[0]=k'1*X'[0]+ k'2*X'[1]+ k'3*X'[2]+ k'4*X'[3]+ k'5*X'[4]+ k'6*X'[5]+ k'7*X'[6]+ k'8*X'[7]$$

DFG of Generic representation of 8-Point IDCT to compute 1st sample:

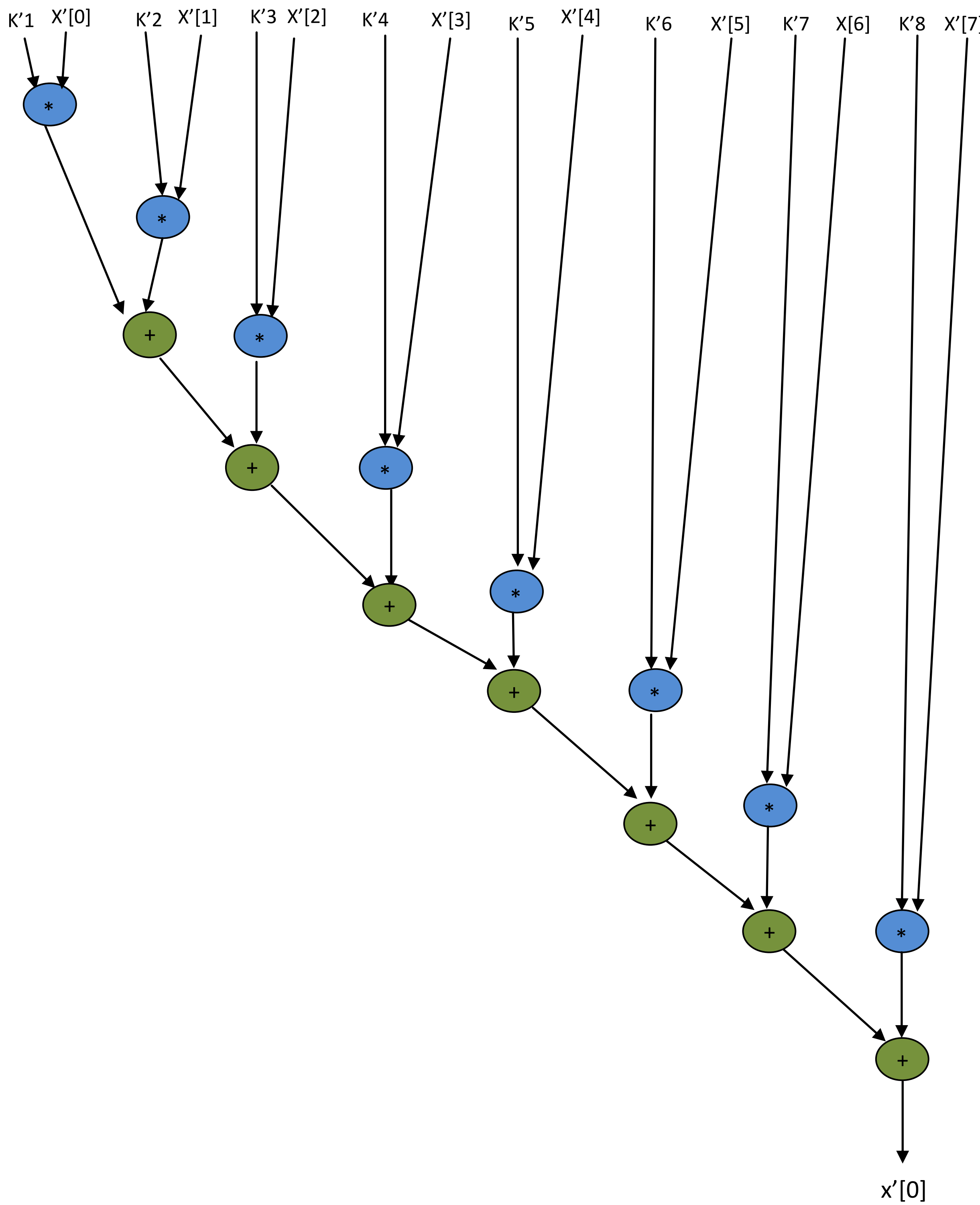


Figure 38

Scheduled DFG of 8-point IDCT based on Resource constraints (1A, 1M).

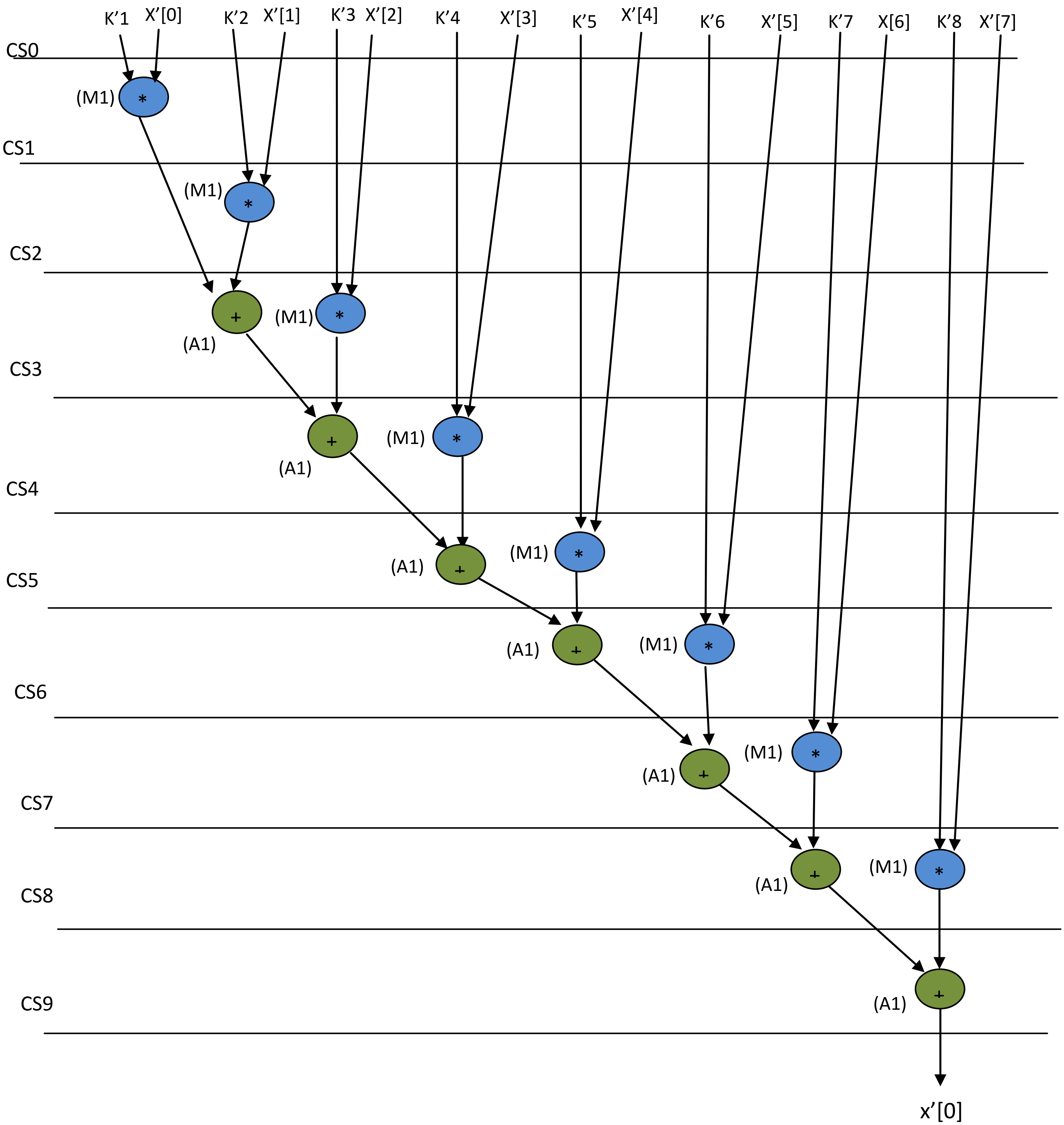


Figure 39

8-Point IDCT: x'[0]=k'1*X'[0]+ k'2*X'[1]+ k'3*X'[2]+ k'4*X'[3]+ k'5*X'[4]+ k'6*X'[5]+ k'7*X'[6]+ k'8*X'[7]. The respective datapath is shown below.

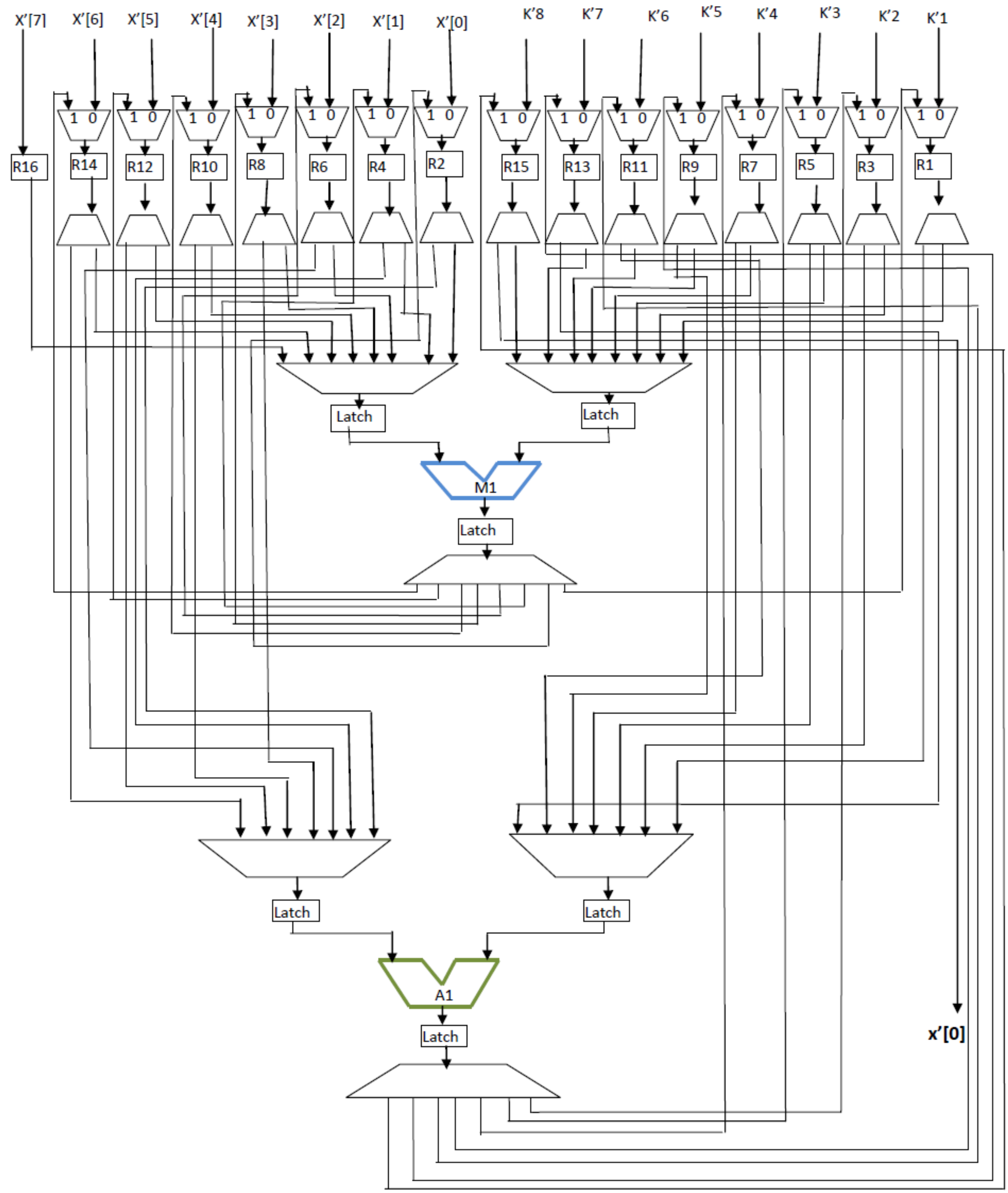


Figure 40

[G] FIR

FIR EQUATION:

Y’[n]=h0*X’[n]+ h1*X’[n-1]+ h2*X’[n-2]+ h3*X’[n-3].

DFG of FIR:

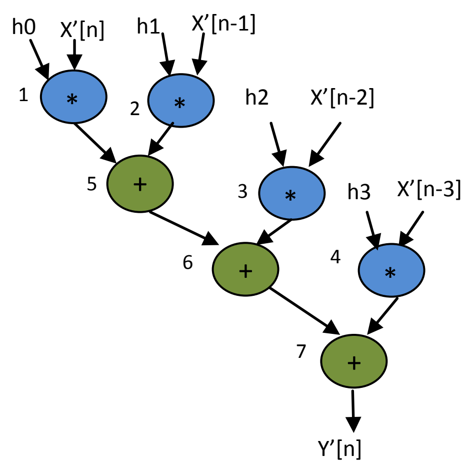


Figure 41

Scheduled DFG of FIR Based on Resource constraints (1A, 1M)

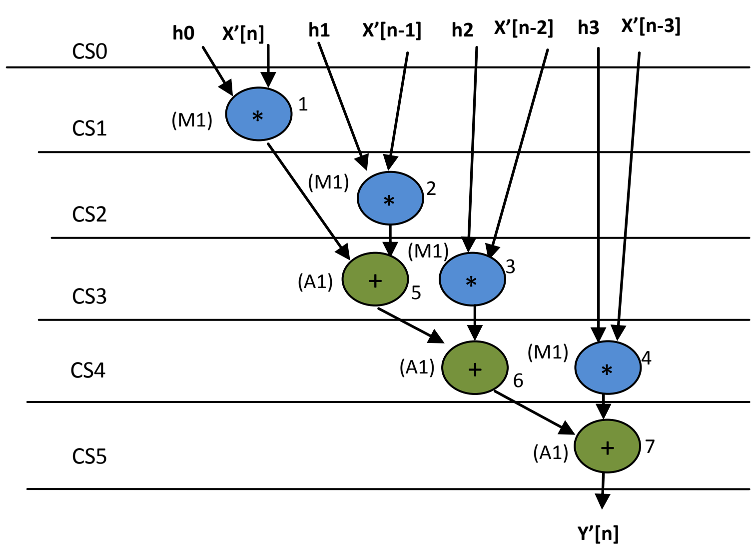


Figure 42

FIR: Y'[n]=h0*X'[n]+ h1*X'[n-1]+ h2*X'[n-2]+ h3*X'[n-3]. The respective datapath is shown below.

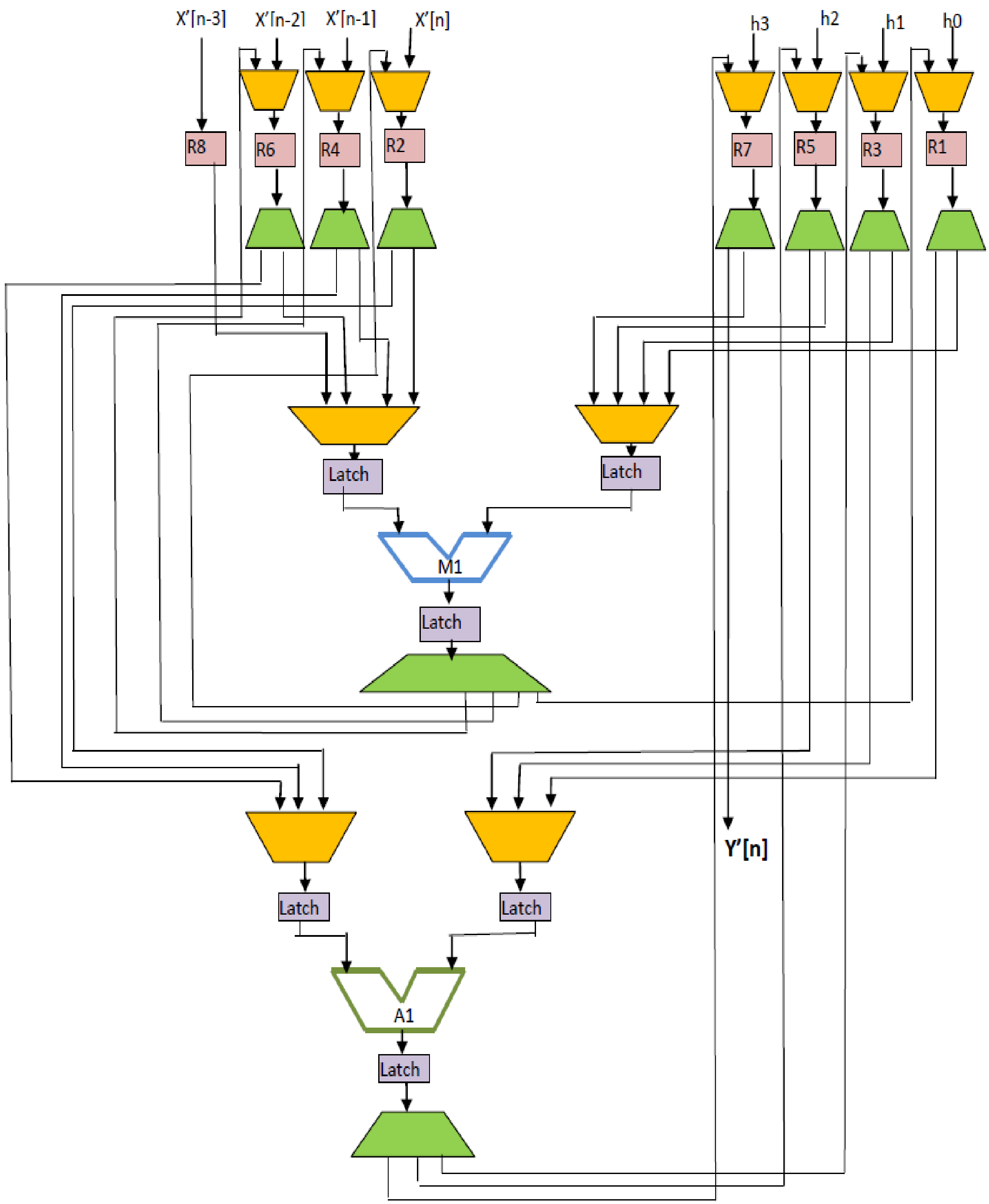


Figure 43

[H] IIR equation:

Y[n]=b0*X[n]+ b1*X[n-1]+ b2*X[n-2]+b3*X[n-3]+(- a1)*Y[n-1]+(- a2)*Y[n-2]+(-a3)*Y[n-3].

DFG of IIR:

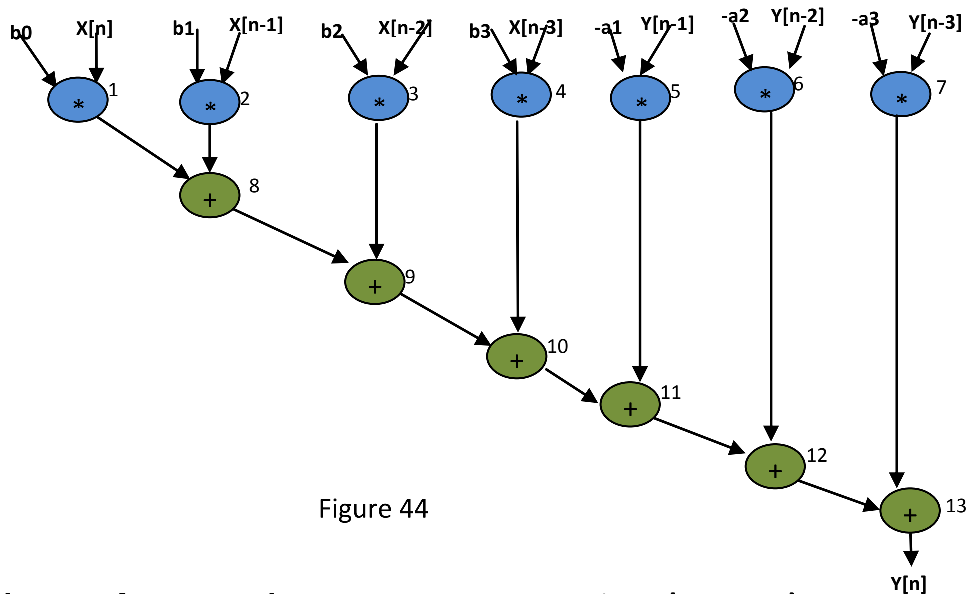
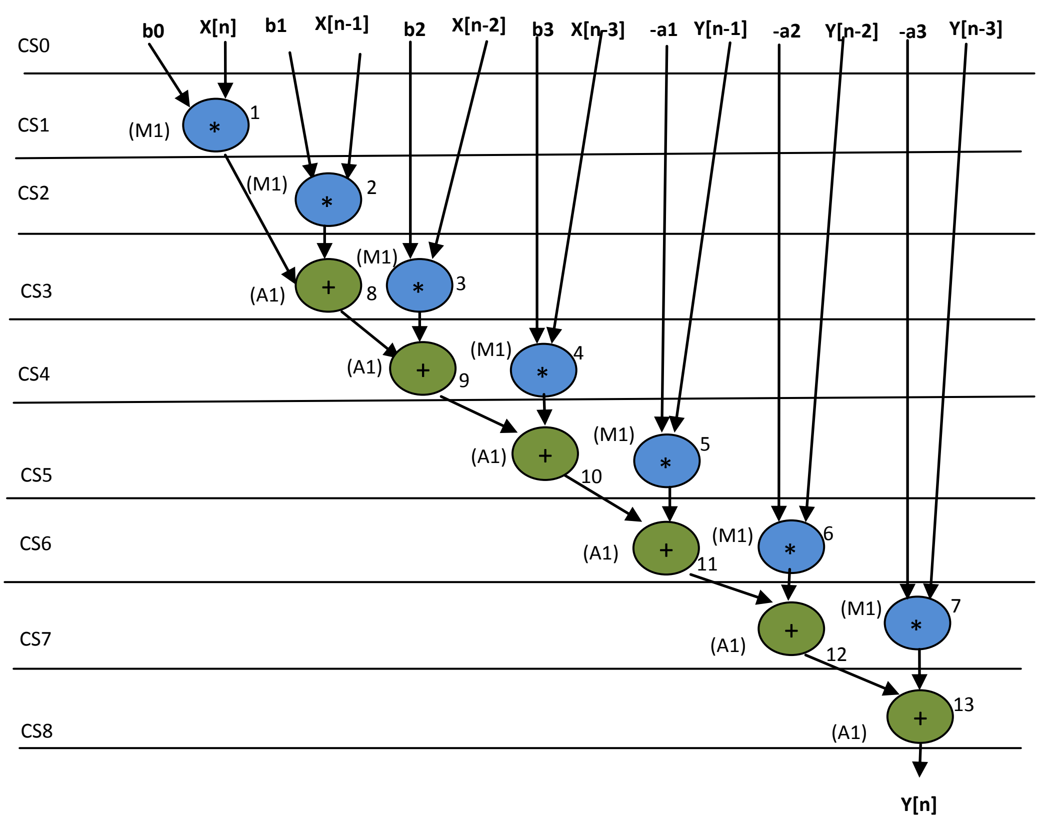


Figure 44

Scheduled DFG of IIR Based on Resource constraints (1A, 1M)



IIR: Y[n]=b0*X[n]+ b1*X[n-1]+ b2*X[n-2]+b3*X[n-3] +(- a1)*Y[n-1]+(- a2)*Y[n-2]+(- a3)*Y[n-3]. The res

Figure 45

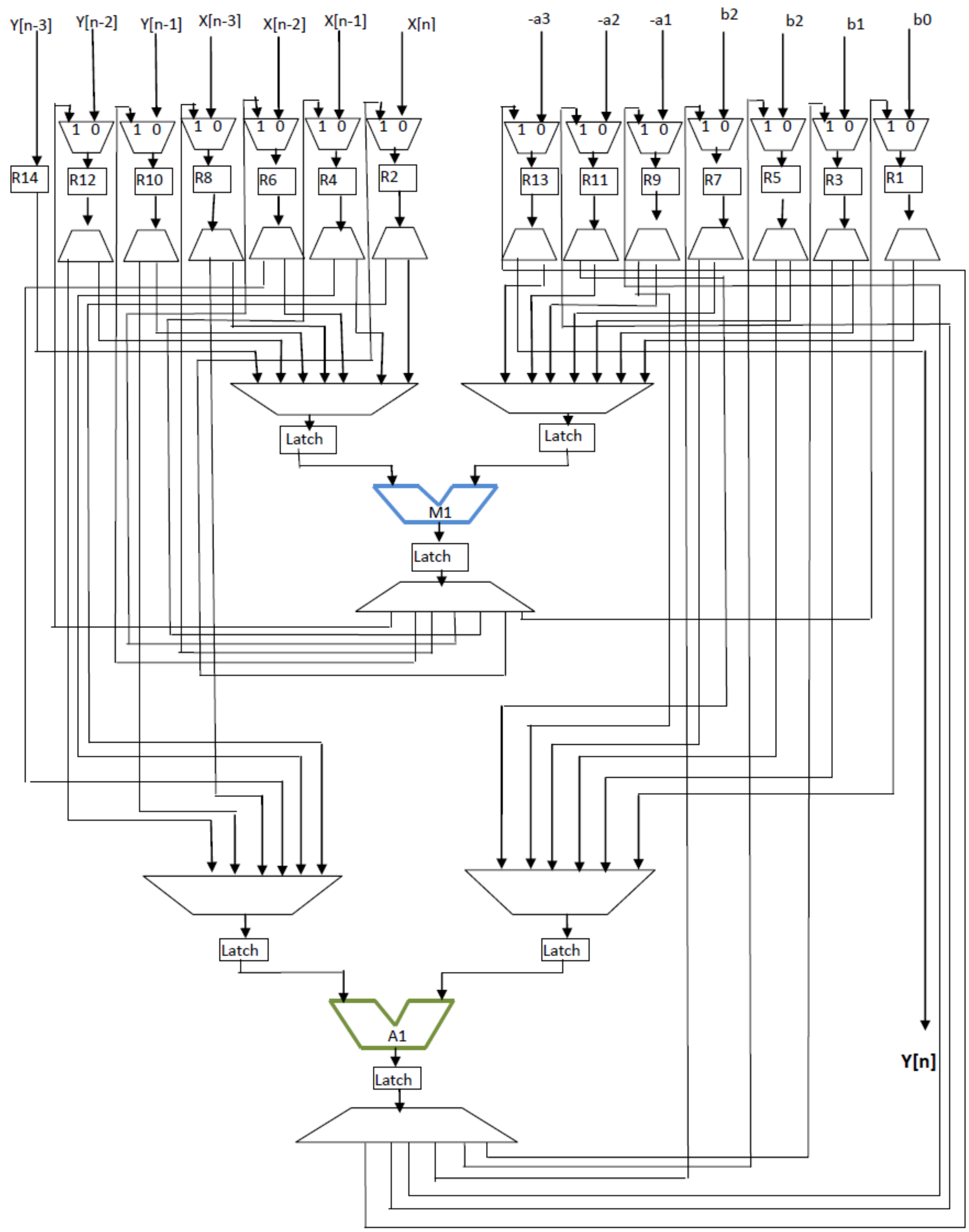

Y[n-3]
Y[n-2]
Y[n-1]
X[n-3]
X[n-2]
X[n-1]
X[n]
-a3
-a2
-a1
b2
b2
b1
b0
R14
R12
R10
R8
R6
R4
R2
R13
R11
R9
R7
R5
R3
R1
Latch
Latch
M1
Latch
Latch
Latch
A1
Latch
Y[n]

Figure 46